

Exceptional collections for canonical stacks of log del Pezzo surfaces with $\frac{1}{3}(1, 1)$ singularities

Alex Junior Gomez Saltachin

Departamento de Matemática, Pontifícia Universidade Católica do Rio de Janeiro
Rua Marquês de São Vicente, 225, Rio de Janeiro, RJ 22451-900, Brazil
alex.gomez@pucp.edu.pe

Abstract

We study canonical smooth Deligne–Mumford stacks and their coarse log del Pezzo surfaces when all singularities are of type $\frac{1}{3}(1, 1)$. Using the Ishii–Ueda special McKay correspondence together with the rationality of the minimal resolution, we obtain full exceptional collections on the canonical stack and the length formula

$$\ell = 12 - K_X^2 + \frac{4r}{3},$$

where r is the number of singular points. We make this construction explicit for the Corti–Heuberger classification: for each of the six qG-rigid cascade bases we give a marked blow-up presentation of its minimal resolution and an explicit Orlov exceptional collection, which then propagates through the chosen smooth-point blow-ups to every index-one member of its cascade. For a general degree-10 hypersurface $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$, we further compute the Euler form of the resulting length-13 collection and show that Karmazyn–Kuznetsov–Shinder descent gives a semiorthogonal decomposition of $D^b(\text{coh } X_{10})$ with a $K(3, 1)$ -component and ten exceptional objects. Finally, we compute the Brauer groups throughout the six cascades: they vanish for one through five singular points, while the six-point cascade has Brauer group $\mathbb{Z}/3$, obstructing untwisted adherent KKS descent.

2020 Mathematics Subject Classification. Primary 14F08; Secondary 14J45, 14A20, 14J17, 14E15, 14J26.

Keywords. Log del Pezzo surfaces, canonical stacks, exceptional collections, semiorthogonal decompositions, quotient singularities, Brauer groups, special McKay correspondence.

1 Introduction

Let X be a log del Pezzo surface over $\mathbb{C}$. Thus X is a normal projective surface with klt singularities and ample anticanonical divisor $-K_X$. In dimension two, klt singularities are precisely quotient singularities. In this paper we focus on the case where all singularities are cyclic quotient singularities of type $\frac{1}{3}(1, 1)$.

We study two related derived categories attached to X . The first is the derived category of the canonical smooth Deligne–Mumford stack $\pi : \mathcal{X} \rightarrow X$. This stack is isomorphic to X over the smooth locus and records the stabilizer groups at the quotient singularities. The second is the derived category $D^b(\text{coh } X)$ of the singular surface

itself. These categories reflect different geometries: $\mathcal{X}$ is smooth as a stack, whereas X is singular as a variety.

Quasismooth well-formed hypersurfaces in weighted projective spaces give a concrete setting for this distinction: the same equation defines a smooth stack and a possibly singular coarse surface. Johnson–Kollár classified the anticanonically embedded log del Pezzo hypersurfaces of this kind in weighted projective 3-spaces [17]. Elagin constructed exceptional collections for a family of non-toric log-terminal del Pezzo hypersurfaces, treated as smooth stacks [11], while Gugiatti–Rota gave explicit full exceptional collections for the canonical stacks of the main Johnson–Kollár series [12]. Here we focus instead on surfaces with only $\frac{1}{3}(1,1)$ -points: the Corti–Heuberger classification allows us to specify initial collections on the resolutions of all six rigid bases and propagate them through chosen marked index-one cascade presentations, while X_{10} provides the worked comparison with the singular coarse surface through KKS descent.

The abstract existence of a full exceptional collection on $D^b(\mathrm{coh} \mathcal{X})$ follows by combining the Ishii–Ueda special McKay correspondence [16, Theorem 1.4] with the full exceptional collection on the minimal resolution supplied by Corollary 2.7. This argument is already recorded in [12, Remark 4.8]. More precisely, Ishii–Ueda construct a fully faithful functor

$$\Phi : D^b(\mathrm{coh} \tilde{X}) \longrightarrow D^b(\mathrm{coh} \mathcal{X}),$$

and identify its semiorthogonal complement in terms of the non-special representations of the local stabilizer groups [16, Proposition 1.1 and Theorem 1.2]. For a singular point p of type $\frac{1}{3}(1,1)$, the unique non-special representation of μ_3 is ρ_2 [16, Theorem 3.2] and [12, Example 2.6]. If $\iota_p : B\mu_3 \hookrightarrow \mathcal{X}$ is the residual gerbe over p , we set

$$\mathcal{E}_p := \iota_{p*}(\mathcal{O}_{B\mu_3} \otimes \rho_2).$$

For singularities of type $\frac{1}{3}(1,1)$, this description makes the local contribution explicit: each stacky point contributes the single exceptional object $\mathcal{E}_p$ supported on its residual gerbe. It also leads directly to the length formula in Theorem 1.1.

The Corti–Heuberger classification by log del Pezzo cascades supplies the geometric framework [7]. It organizes the relevant surfaces into sequences of blow-ups from $\mathbb{Q}$ -Gorenstein rigid (qG -rigid) bases. Once a full exceptional collection is fixed on the minimal resolution of such a base, Orlov’s blow-up formula and the local Ishii–Ueda contribution carry it along the cascade. The cascades therefore control both the birational geometry and the passage to the stack-theoretic collections.

This abstract argument leaves the initial collection on each rigid resolution unspecified. Proposition 5.4 and Appendix B supply explicit marked blow-up models and Orlov collections for all six bases, making the construction explicit once a marked cascade presentation is chosen.

For orientation, $\tilde{X}$ denotes the resolution and $\mathcal{X}$ the canonical stack; N and ℓ count the objects in their respective collections, and $\mathcal{E}_p$ denotes the local stack object at p . In the cascade notation, $X_{k,d}$ denotes an index-one family with k singular points and degree d , and Z_k is its rigid base.

Specializing this formal mechanism to $\frac{1}{3}(1,1)$ -points gives the following explicit description and length formula.

Theorem 1.1. *Let X be a log del Pezzo surface whose singularities are $p_1, \dots, p_r$, all of type $\frac{1}{3}(1,1)$. Let $\mathcal{X}$ be its canonical smooth Deligne–Mumford stack, and let $\tilde{X} \rightarrow X$ be the minimal resolution. The category $D^b(\mathrm{coh} \mathcal{X})$ admits a full exceptional collection*

of length

$$\ell = 12 - K_X^2 + \frac{4r}{3}.$$

More precisely, let

$$\langle G_1, \dots, G_N \rangle$$

be a full exceptional collection on $D^b(\mathrm{coh} \tilde{X})$, where

$$N = \mathrm{rk} K_0(\tilde{X}) = 12 - K_X^2 + \frac{r}{3}.$$

If $\Phi : D^b(\mathrm{coh} \tilde{X}) \rightarrow D^b(\mathrm{coh} \mathcal{X})$ is the Ishii–Ueda fully faithful functor, the ordered collection

$$\langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_r}, \Phi(G_1), \dots, \Phi(G_N) \rangle$$

is a full exceptional collection on $D^b(\mathrm{coh} \mathcal{X})$, where

$$\mathcal{E}_{p_i} = \iota_{p_i*}(\mathcal{O}_{B\mu_3} \otimes \rho_2), \quad \iota_{p_i} : B\mu_3 \hookrightarrow \mathcal{X}$$

is the inclusion of the residual gerbe over p_i , and ρ_2 is the unique non-special representation of μ_3 .

The theorem is established in [Section 3](#) by combining the Ishii–Ueda decomposition with the geometry of the minimal resolution. Its explicit content is the identification of the local object and the length formula: the resolution contributes $12 - K_X^2 + \frac{r}{3}$ objects, and the stack contributes one object at each singular point. The analogous formula for singularities of type $\frac{1}{k}(1, 1)$ for general k , together with a closed Euler-matrix formula in the one-point case, appears in [Section 6](#).

The principal worked example, where we additionally compute the Euler form and carry out KKS descent on the singular coarse surface, is the family of degree-10 hypersurfaces

$$X_{10} = V(f_{10}) \subset \mathbb{P}(1, 2, 3, 5).$$

A general quasismooth well-formed member has a unique singular point of the type considered above and belongs to the Corti–Heuberger family $X_{1,1/3}$. By the *one-point cascade* we mean the cascade containing surfaces with one $\frac{1}{3}(1, 1)$ -point. Corti–Heuberger show that every member of $X_{1,1/3}$ admits a presentation by eight blow-ups of nonsingular points of the qG -rigid base $\mathbb{P}(1, 1, 3)$ [[7](#), Corollary 8(1)]. The Reid–Suzuki construction realizes the general member as the degree-10 anticanonical model of such a blow-up [[23](#), Section 1 and Table 1.1]. On minimal resolutions, this gives eight blow-ups of $\mathbb{F}_3$, all away from its negative section.

This geometry permits an explicit treatment of X_{10} . We display a full exceptional collection of length 13 on its canonical stack, compute its Euler form, and verify the compatibility and adherence conditions required for Karmazyn–Kuznetsov–Shinder descent to the singular surface [[18](#)].

For a blow-up $\beta_i : Y_i \rightarrow Y_{i-1}$ with exceptional curve D_i , Orlov’s blow-up formula contributes $\mathcal{O}_{D_i}(-1)$. We denote its image after pullback through the subsequent blow-ups by $G_i \in D^b(\mathrm{coh} \tilde{X}_{10})$, and retain the notation $\mathcal{E}_p$ and Φ introduced above.

Theorem 1.2. *Let $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ be a general member of the family of degree-10 log del Pezzo hypersurfaces. Its unique singularity p is of type $\frac{1}{3}(1, 1)$, and X_{10} is the anticanonical model of the blow-up of $\mathbb{P}(1, 1, 3)$ at eight general smooth points.*

On minimal resolutions, this gives a sequence

$$Y_8 = \tilde{X}_{10} \xrightarrow{\beta_8} Y_7 \rightarrow \cdots \rightarrow Y_1 \xrightarrow{\beta_1} Y_0 = \mathbb{F}_3,$$

where each β_i is the blow-up of a point away from the negative section $C_0 \subset \mathbb{F}_3$ and its successive strict transforms. Let f be the fiber class on $\mathbb{F}_3$, and let $\sigma : \tilde{X}_{10} \rightarrow \mathbb{F}_3$ be the composition of the β_i . The category $D^b(\text{coh } \mathcal{X}_{10})$ has the following full exceptional collection of length 13:

$$\langle \mathcal{E}_p, \Phi(G_8), \dots, \Phi(G_1), \Phi\sigma^*\mathcal{O}, \Phi\sigma^*\mathcal{O}(f), \Phi\sigma^*\mathcal{O}(C_0 + 3f), \Phi\sigma^*\mathcal{O}(C_0 + 4f) \rangle.$$

The singular surface X_{10} also admits a semiorthogonal decomposition

$$D^b(\text{coh } X_{10}) = \langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{10} \rangle,$$

where $F_1, \dots, F_{10}$ are exceptional objects and

$$K(3, 1) \cong \mathbb{C}[z_1, z_2]/(z_1, z_2)^2.$$

The geometry and categorical constructions in [Theorem 1.2](#) are developed in [Section 5](#), including the exceptional collection, its Euler form, the vanishing of the Brauer group, and the KKS descent to the singular surface. Beyond this worked example, the explicit collections extend to all six rigid bases and their index-one cascades.

Theorem 1.3 (Explicit collections in the Corti–Heuberger cascades). *For each rigid base Z_k , $1 \leq k \leq 6$, the minimal resolution $\tilde{Z}_k$ admits the explicit marked blow-up presentation constructed in [Proposition 5.4](#) and [Appendix B](#), with associated explicit Orlov full exceptional collection $\mathcal{E}_k = \langle A_1^{(k)}, \dots, A_{N_k}^{(k)} \rangle$.*

Let $X \in X_{k,d}$ be an index-one member of the k -th cascade, and choose a marked Corti–Heuberger presentation $X \rightarrow Z_k$ by $s = K_{Z_k}^2 - d$ smooth-point blow-ups. Let $\gamma : \tilde{X} \rightarrow \tilde{Z}_k$ be the induced map of minimal resolutions, and let $H_1, \dots, H_s$ be the corresponding Orlov objects, pulled back in the derived sense from their creation stages. Then $\langle H_s, \dots, H_1, L\gamma^\mathcal{E}_k \rangle$ is an explicitly ordered full exceptional collection on $D^b(\text{coh } \tilde{X})$. On the canonical stack $\mathcal{X}$, the Ishii–Ueda embedding Φ gives the full exceptional collection*

$$\langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_k}, \Phi(H_s), \dots, \Phi(H_1), \Phi(L\gamma^*\mathcal{E}_k) \rangle,$$

where the p_i are the singular points and functors act entry by entry on the ordered base collection. The construction is explicit relative to the chosen marked presentation; no canonical choice of presentation is asserted.

The base constructions and [Corollary B.3](#) prove [Theorem 1.3](#). For the four-point base, contracting a floating (-1) -curve in the quadrilateral model is incorporated into an elementary transformation of $\mathbb{F}_1$, yielding the seven-point blow-up presentation from $\mathbb{F}_2$ of [Proposition B.2](#).

Different marked presentations may produce different ordered collections. We do not establish whether collections arising from different presentations of the same surface lie in a single mutation orbit; [Theorem 1.3](#) asserts neither uniqueness nor independence of these choices up to mutation.

On the singular coarse-surface side, the vanishing of the Brauer group for X_{10} leads to the corresponding question throughout the six cascades, which [Appendix A](#) answers uniformly.

Theorem 1.4 (Brauer obstruction in the Corti–Heuberger cascades). *Let X belong to a Corti–Heuberger cascade with k singular points, all of type $\frac{1}{3}(1, 1)$. Its Brauer group is*

$$\mathrm{Br}(X) \cong \begin{cases} 0, & 1 \leq k \leq 5, \\ \mathbb{Z}/3, & k = 6. \end{cases}$$

In particular, among the six Corti–Heuberger cascades, the six-point cascade is the unique one for which the Brauer group obstructs an untwisted adherent decomposition in the sense of Karmazyn–Kuznetsov–Shinder.

The canonical stack of Z_6 still has the explicit full exceptional collection of [Theorem 1.3](#): the nonzero Brauer group obstructs untwisted adherent KKS descent on the singular coarse surface, not the stack collection. It does not rule out unrelated semiorthogonal decompositions or twisted adherent constructions.

The paper is organized as follows. [Section 2](#) treats resolutions and exceptional collections; [Section 3](#) introduces canonical stacks and the special McKay correspondence; and [Section 4](#) concerns the singular surface and KKS descent. [Section 5](#) develops the Corti–Heuberger cascades and the X_{10} example, including its Euler form. The general $\frac{1}{k}(1, 1)$ length formula and the closed Euler-matrix formula in the one-point case appear in [Section 6](#). [Appendix A](#) contains the Brauer-group and exceptional-lattice calculations and the KKS obstruction. [Appendix B](#) constructs explicit marked blow-up models and full exceptional collections for the six rigid Corti–Heuberger base resolutions and derives the resulting explicit collections throughout the index-one cascades, proving [Theorem 1.3](#).

Remark 1.5 (Notation guide). The geometric curve labels distinguish roles in the chosen presentation:

E, E_i	curves contracted to quotient singularities,
D, D_i	exceptional curves of smooth point blow-ups,
C_0	the negative section of a Hirzebruch surface $\mathbb{F}_n$.

Strict transforms and total pullbacks are specified when needed. The calligraphic symbol $\mathcal{E}_p$ denotes the exceptional *object* supported on the residual gerbe over p , not a curve; $\mathrm{Ext}^\bullet$ and χ denote Ext groups and the Euler pairing.

For the categorical constructions, we use the following dictionary:

- $\Phi : D^b(\mathrm{coh} \tilde{X}) \rightarrow D^b(\mathrm{coh} \mathcal{X})$ is the Ishii–Ueda embedding, and Ψ is its left adjoint, used in the Euler-form computations. The subscript in Φ_k specifies the rigid base Z_k .
- In a marked blow-up construction, G_i is the Orlov object $\mathcal{O}_{D_i}(-1)$ pulled back in the derived sense through all subsequent blow-ups, as in (5); it need not be a sheaf on the final strict transform of D_i . In the abstract statement of [Theorem 1.1](#), the G_i instead denote arbitrary entries of a full exceptional collection on the resolution.
- $\mathcal{E}_k = \langle A_1^{(k)}, \dots, A_{N_k}^{(k)} \rangle$ is the initial collection on $\tilde{Z}_k$; the $A_i^{(k)}$ range over all its entries, including the pulled-back starting line bundles. The H_i are the Orlov objects of the additional blow-ups along a chosen cascade, with the same derived-pullback convention.

Applying a functor to an ordered collection means applying it entry by entry without changing the order.

2 Resolutions and exceptional collections

Rationality and Orlov’s blow-up formula provide the starting collection for the stack construction in [Section 3](#). Rationality expresses a log del Pezzo minimal resolution as a sequence of point blow-ups of $\mathbb{P}^2$ or a Hirzebruch surface; Orlov’s formula transports their standard full exceptional collections to the resolution. Together these ingredients yield [Corollary 2.7](#). All algebraic varieties are defined over $\mathbb{C}$.

For a birational morphism of surfaces, we use *exceptional curve* for an irreducible curve contracted by the morphism and *exceptional locus* for the union of all such curves. The exceptional locus of a point blow-up is a single curve.

Standard background on log del Pezzo surfaces is developed in [2, section “Basic facts about log del Pezzo surfaces”]. Rationality of the minimal resolution is recalled in [12, Remark 4.3]. For exceptional collections on smooth rational surfaces, we follow [14].

Definition 2.1. A *log del Pezzo surface* is a normal projective surface X with klt singularities such that $-K_X$ is ample.

For normal surfaces, klt singularities are precisely quotient singularities [19, Corollary 5.21]. They are also rational [19, Theorem 5.22]. Hence, if $f : \tilde{X} \rightarrow X$ is any resolution, then $f_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ and $R^if_*\mathcal{O}_{\tilde{X}} = 0$ for all $i > 0$. The Leray spectral sequence therefore gives a canonical isomorphism

$$H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) \cong H^1(X, \mathcal{O}_X).$$

For completeness, we recall the rationality argument, which also fixes the discrepancy convention used below.

Proposition 2.2. *Let X be a log del Pezzo surface, and let $f : \tilde{X} \rightarrow X$ be its minimal resolution. The surface $\tilde{X}$ is smooth and rational. In particular, X is rational.*

Proof. Since X has klt singularities and $-K_X$ is ample, Kawamata–Viehweg vanishing gives $H^1(X, \mathcal{O}_X) = 0$. Since the singularities of X are rational, the Leray isomorphism above gives $H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$.

It remains to show that $H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(2K_{\tilde{X}})) = 0$. Suppose, to the contrary, that an effective divisor $D \sim 2K_{\tilde{X}}$ exists. The pullback $-f^*K_X$ is nef and big because $-K_X$ is ample, and hence $(-f^*K_X) \cdot D \geq 0$. On the other hand, write the discrepancy formula as

$$K_{\tilde{X}} = f^*K_X + \sum_i a_i E_i$$

. Since $f^*K_X \cdot E_i = 0$ for every f -exceptional curve E_i , this formula gives

$$(-f^*K_X) \cdot D = (-f^*K_X) \cdot 2K_{\tilde{X}} = -2K_X^2 < 0.$$

This contradiction implies $H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(2K_{\tilde{X}})) = 0$. Thus, by Castelnuovo’s rationality criterion, $\tilde{X}$ is rational. Since X is birational to $\tilde{X}$, the surface X is rational as well. $\square$

Example 2.3. The weighted projective plane $\mathbb{P}(1, 1, 3)$ is a log del Pezzo surface with a unique singularity of type $\frac{1}{3}(1, 1)$. Its minimal resolution is the Hirzebruch surface $\mathbb{F}_3 = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(3))$, and its exceptional locus is the negative section $C_0 \subset \mathbb{F}_3$, with $C_0^2 = -3$ [18, Example 3.17]. This description also follows from the toric resolution procedure of [9, Chapters 10–11]. This same model reappears in [Section 5](#) as the one-point cascade base $Z_1 = \mathbb{P}(1, 1, 3)$, with resolution $\tilde{Z}_1 = \mathbb{F}_3$.

We now choose the standard collections on the starting smooth surfaces and explain how they pass through point blow-ups.

Definition 2.4. Let $\mathcal{T}$ be a triangulated category. An object $G \in \mathcal{T}$ is *exceptional* if $\mathrm{Hom}(G, G) = \mathbb{C}$ and $\mathrm{Ext}^k(G, G) = 0$ for all $k \neq 0$. An ordered sequence $\langle G_1, \dots, G_n \rangle$ is an *exceptional collection* if each G_i is exceptional and $\mathrm{Ext}^\bullet(G_j, G_i) = 0$ for all $j > i$. The collection is *full* if it generates $\mathcal{T}$ as a triangulated category, that is, under shifts and cones.

The projective line has the standard full exceptional collection

$$D^b(\mathrm{coh} \mathbb{P}^1) = \langle \mathcal{O}, \mathcal{O}(1) \rangle,$$

whereas the projective plane has Beilinson's full exceptional collection [3]

$$D^b(\mathrm{coh} \mathbb{P}^2) = \langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2) \rangle.$$

Orlov's projective bundle formula [22, Corollary 2.7], applied to

$$\mathbb{F}_n = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(n))$$

together with the collection on $\mathbb{P}^1$, gives a full exceptional collection on $\mathbb{F}_n$. Let C_0 denote the negative section and f the fiber class. With the convention $C_0^2 = -n$, $C_0 \cdot f = 1$, and $f^2 = 0$, the relative tautological line bundle is

$$\mathcal{O}_{\mathbb{F}_n}(1) \simeq \mathcal{O}(C_0 + nf).$$

After twisting the projective-bundle decomposition by $\mathcal{O}_{\mathbb{F}_n}(1)$, its two blocks are ordered as

$$D^b(\mathrm{coh} \mathbb{F}_n) = \langle p^* D^b(\mathrm{coh} \mathbb{P}^1), p^* D^b(\mathrm{coh} \mathbb{P}^1) \otimes \mathcal{O}(C_0 + nf) \rangle,$$

where $p : \mathbb{F}_n \rightarrow \mathbb{P}^1$ is the ruling. Expanding the two blocks in this order gives the full exceptional collection

$$\langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + nf), \mathcal{O}(C_0 + (n+1)f) \rangle.$$

Theorem 2.5 (Orlov's blow-up formula [22, Theorem 4.3 and Corollary 4.4]). *Let S be a smooth projective surface, and let $\beta : \mathrm{Bl}_p S \rightarrow S$ be the blow-up of S at a point p , with exceptional curve D . There is a semiorthogonal decomposition*

$$D^b(\mathrm{coh} \mathrm{Bl}_p S) = \langle \mathcal{O}_D(-1), \beta^* D^b(\mathrm{coh} S) \rangle,$$

where $\mathcal{O}_D(-1)$ is regarded as an object of $D^b(\mathrm{coh} \mathrm{Bl}_p S)$ via the inclusion $D \hookrightarrow \mathrm{Bl}_p S$. In particular, if $D^b(\mathrm{coh} S)$ admits a full exceptional collection, the category $D^b(\mathrm{coh} \mathrm{Bl}_p S)$ also admits one.

Corollary 2.6 ([14, Theorem 5.6]). *Every smooth projective rational surface admits a full exceptional collection.*

Proof. Every smooth projective rational surface is obtained from either $\mathbb{P}^2$ or a Hirzebruch surface $\mathbb{F}_n$ by a sequence of blow-ups at points. We induct on the number of blow-ups: the starting surfaces $\mathbb{P}^2$ and $\mathbb{F}_n$ have full exceptional collections, and [Theorem 2.5](#) supplies the induction step by adjoining one exceptional object at each point blow-up. $\square$

Together, [Proposition 2.2](#) and [Corollary 2.6](#) yield the following consequence.

Corollary 2.7. *Let X be a log del Pezzo surface, and let $f : \tilde{X} \rightarrow X$ be its minimal resolution. The category $D^b(\mathrm{coh} \tilde{X})$ admits a full exceptional collection.*

3 Canonical stacks and exceptional collections

This section introduces the stack-theoretic replacement of a singular surface and the local representation-theoretic input for the semiorthogonal decomposition. These ingredients yield the full exceptional collection theorem for canonical stacks of log del Pezzo surfaces with $\frac{1}{3}(1, 1)$ singularities.

Definition 3.1. Let X be a normal surface with quotient singularities. The *canonical stack* of X is the smooth Deligne–Mumford stack $\pi : \mathcal{X} \rightarrow X$ whose coarse moduli space is X , whose local models at quotient singularities are the corresponding quotient stacks, and whose structure morphism is an isomorphism over the smooth locus of X .

At a singular point of type $\frac{1}{3}(1, 1)$, the local analytic model is $\mathbb{A}^2/\mu_3$, where μ_3 acts by $\zeta \cdot (x, y) = (\zeta x, \zeta y)$. Hence the canonical stack is locally $[\mathbb{A}^2/\mu_3]$. Moreover, the minimal resolution of this singularity is determined by the Hirzebruch–Jung continued fraction $\frac{3}{1} = [3]$, so the exceptional locus consists of one smooth rational curve E with $E^2 = -3$.

For the local terminology of the special McKay correspondence, let $G \subset \mathrm{GL}_2(\mathbb{C})$ be a finite small subgroup, let $R = \mathbb{C}[x, y]$, and let

$$\tau : Y \longrightarrow \mathrm{Spec} R^G$$

be the minimal resolution of the quotient singularity. Here small means that G contains no pseudo-reflections. For an irreducible representation ρ of G , set

$$M_\rho := (R \otimes \rho^\vee)^G.$$

This is a reflexive R^G -module. The associated *full sheaf* on Y is

$$\mathcal{M}_\rho := \tau^* M_\rho / \text{torsion}.$$

Wunram’s correspondence associates indecomposable full sheaves on Y with irreducible representations of G . A full sheaf $\mathcal{M}_\rho$ is called *special* if

$$H^1(Y, \mathcal{M}_\rho^\vee) = 0.$$

Equivalently, the corresponding representation ρ is called a *special representation*. A representation is called *non-special* if it is not special. This terminology follows Wunram [25] and the exposition in [16, Section 2].

Lemma 3.2. *For a cyclic quotient singularity of type $\frac{1}{3}(1, 1)$, the unique non-special representation of the stabilizer group μ_3 is ρ_2 , where $\rho_i(\zeta) = \zeta^i$.*

Proof. In the cyclic case $\frac{1}{n}(1, q)$, the special representations can be read from the sequence associated with the Hirzebruch–Jung continued fraction. If $\frac{n}{q} = [b_1, \dots, b_s]$, this sequence begins with $i_0 = n$ and $i_1 = q$ and is defined recursively by $i_{t+1} = b_t i_t - i_{t-1}$; the entries preceding the terminal zero index the special representations. This is Wunram’s cyclic special McKay correspondence: the continued-fraction labels $i_1, \dots, i_s$ identify the nontrivial special representations associated with the resolution curves of squares $-b_1, \dots, -b_s$, while i_0 labels the trivial representation [16, Theorems 2.3 and 3.2]. In our case, $n = 3$, $q = 1$, and $\frac{3}{1} = [3]$, so the sequence is

$$i_0 = 3, \quad i_1 = 1, \quad i_2 = 0.$$

Character indices are read modulo 3. The special representations are therefore

$$\rho_{i_0} = \rho_3 = \rho_0, \quad \rho_{i_1} = \rho_1.$$

Since the irreducible characters of μ_3 are exactly ρ_0, ρ_1, ρ_2 , the unique non-special representation is ρ_2 . This local computation also appears in [12, Example 2.6]. $\square$

Definition 3.3. Let $p \in X$ be a singular point of type $\frac{1}{3}(1, 1)$, and let

$$\iota_p : B\mu_3 \hookrightarrow \mathcal{X}$$

be the residual gerbe over p . The *local exceptional object* associated with p is

$$\mathcal{E}_p := \iota_{p*}(\mathcal{O}_{B\mu_3} \otimes \rho_2).$$

Equivalently, after identifying an étale neighborhood of p with the quotient stack $[\mathbb{A}^2/\mu_3]$, the object $\mathcal{E}_p$ is represented by the μ_3 -equivariant module

$$\mathbb{C}[x, y]/(x, y) \otimes \rho_2.$$

In the local notation of Ishii–Ueda, this is the object $\mathcal{O}_0 \otimes \rho_2$.

Theorem 3.4 (Ishii–Ueda [16, Theorem 1.4]). *Let X be a surface with at worst quotient singularities, let $\mathcal{X}$ be its canonical stack, and let $Y \rightarrow X$ be the minimal resolution. There is a fully faithful functor*

$$\Phi : D^b(\mathrm{coh} Y) \rightarrow D^b(\mathrm{coh} \mathcal{X})$$

and a semiorthogonal decomposition

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{E}_1, \dots, \mathcal{E}_\ell, \Phi(D^b(\mathrm{coh} Y)) \rangle,$$

where $\mathcal{E}_1, \dots, \mathcal{E}_\ell$ form an exceptional collection.

For the cyclic singularities $\frac{1}{n}(1, 1)$ used below, the exceptional collection in the Ishii–Ueda complement is given by the residual-gerbe simple objects corresponding to the non-special representations [16, Proposition 1.1 and Theorem 1.2] and [12, Example 2.6]. Hence a singularity of type $\frac{1}{3}(1, 1)$ contributes exactly one exceptional object to the complement, namely the object $\mathcal{E}_p$ defined above.

These local objects give the desired global decomposition.

Theorem 3.5. *Let X be a log del Pezzo surface whose singularities are all of type $\frac{1}{3}(1, 1)$. Let $\mathcal{X}$ be its canonical stack, and let $\tilde{X} \rightarrow X$ be the minimal resolution. Suppose that the singular points of X are $p_1, \dots, p_r$. There is a semiorthogonal decomposition*

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_r}, \Phi(D^b(\mathrm{coh} \tilde{X})) \rangle.$$

In particular, $D^b(\mathrm{coh} \mathcal{X})$ admits a full exceptional collection.

Proof. By Proposition 2.2, the minimal resolution $\tilde{X}$ is a smooth rational surface. Therefore $D^b(\mathrm{coh} \tilde{X})$ admits a full exceptional collection by Corollary 2.6.

By Theorem 3.4, there is a fully faithful functor

$$\Phi : D^b(\mathrm{coh} \tilde{X}) \hookrightarrow D^b(\mathrm{coh} \mathcal{X})$$

and a semiorthogonal decomposition whose complement is generated by the local exceptional objects associated with the non-special representations. Since each singular point p_i has type $\frac{1}{3}(1, 1)$, its unique non-special representation is ρ_2 by [Lemma 3.2](#). Hence the local exceptional collection over p_i consists of the single object

$$\mathcal{E}_{p_i} = \iota_{p_i*}(\mathcal{O}_{B\mu_3} \otimes \rho_2).$$

Accordingly, the decomposition is

$$D^b(\text{coh } \mathcal{X}) = \langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_r}, \Phi(D^b(\text{coh } \tilde{X})) \rangle.$$

The objects $\mathcal{E}_{p_i}$ have disjoint supports for distinct p_i , so they are mutually orthogonal. Since $D^b(\text{coh } \tilde{X})$ has a full exceptional collection and the remaining components are generated by the exceptional objects $\mathcal{E}_{p_i}$, the category $D^b(\text{coh } \mathcal{X})$ has a full exceptional collection. $\square$

Corollary 3.6. *With the notation of [Theorem 3.5](#), the category $D^b(\text{coh } \mathcal{X})$ has a full exceptional collection of length*

$$12 - K_X^2 + \frac{4r}{3}.$$

More precisely, the minimal resolution satisfies

$$\text{rk } K_0(\tilde{X}) = 12 - K_X^2 + \frac{r}{3},$$

and the canonical stack adds one further exceptional object for each singular point.

Proof. Write $f : \tilde{X} \rightarrow X$ for the minimal resolution and E_i for the curve over p_i . As described above, these are disjoint smooth rational curves of square -3 . Adjunction gives $K_{\tilde{X}} \cdot E_i = -2 - E_i^2 = 1$; since $f^*K_X \cdot E_i = 0$, each discrepancy coefficient is $1/(-3) = -1/3$. Thus

$$K_{\tilde{X}} = f^*K_X - \frac{1}{3} \sum_{i=1}^r E_i.$$

Squaring this identity gives

$$K_{\tilde{X}}^2 = K_X^2 + \frac{1}{9} \sum_{i=1}^r E_i^2 = K_X^2 - \frac{r}{3}.$$

For the smooth rational surface $\tilde{X}$, the relation $\rho(\tilde{X}) = 10 - K_{\tilde{X}}^2$ shows that the Grothendieck-group rank is

$$\text{rk } K_0(\tilde{X}) = 2 + \rho(\tilde{X}) = 12 - K_{\tilde{X}}^2 = 12 - K_X^2 + \frac{r}{3}.$$

Finally, the semiorthogonal decomposition in [Theorem 3.5](#) adds one exceptional object $\mathcal{E}_{p_i}$ for each singular point p_i . Thus the full exceptional collection on $D^b(\text{coh } \mathcal{X})$ has length

$$\left(12 - K_X^2 + \frac{r}{3}\right) + r = 12 - K_X^2 + \frac{4r}{3}. \quad \square$$

Together with [Theorem 3.5](#), this length computation completes the proof of [Theorem 1.1](#).

4 The singular surface and the KKS approach

We next pass from the canonical stack to the singular surface X . The strategy is to construct a decomposition on $\mathbb{F}_3$ that is compatible with the contraction of its negative section, to propagate both compatibility and untwisted adherence through blow-ups away from that section, and finally to apply the Karmazyn–Kuznetsov–Shinder (KKS) descent and identification theorems.

The stack-theoretic collection does not by itself provide a decomposition of $D^b(\mathrm{coh} X)$; the passage to the coarse surface requires a separate descent argument. The KKS descent method applies to surfaces with rational singularities, a hypothesis satisfied here by the klt property, as recalled in [Section 2](#).

We first recall the compatibility condition in the resolution setting used throughout this section.

Definition 4.1. Let $\tau : \tilde{X} \rightarrow X$ be a resolution of a normal projective surface, viewed as a birational contraction, and let

$$D^b(\mathrm{coh} \tilde{X}) = \langle \tilde{\mathcal{A}}_1, \dots, \tilde{\mathcal{A}}_n \rangle$$

be a semiorthogonal decomposition. The decomposition is *compatible with the contraction* τ if, for every irreducible curve E contracted by τ , the object $\mathcal{O}_E(-1)$ belongs to one of the components $\tilde{\mathcal{A}}_i$ [[18](#), Definition 2.7].

For a decomposition compatible with τ , the descent theorem [[18](#), Theorem 2.12] produces a semiorthogonal decomposition of $D^b(\mathrm{coh} X)$. Compatibility is therefore the condition required for descent; the stronger adherence condition serves instead to identify one of the descended components.

In the one-curve case used in this paper, untwisted adherence has the following concrete meaning.

Definition 4.2. Let S be a smooth projective surface such that $H^\bullet(S, \mathcal{O}_S) \cong \mathbb{C}$, and let $E \subset S$ be a smooth rational curve with $E^2 = -r$, where $r \geq 2$. A triangulated subcategory $\mathcal{A} \subset D^b(\mathrm{coh} S)$ is *untwisted adherent* to E if there is a line bundle L_0 such that

$$\mathcal{A} = \langle L_0, \mathcal{O}_E(-1) \rangle, \quad L_0 \cdot E = r - 1.$$

By [[18](#), Lemma 3.5, Definition 3.6, and Example 3.7], this is equivalent to a presentation

$$\mathcal{A} = \langle L_0, L_1 \rangle, \quad L_1 = L_0(E), \quad L_0 \cdot E = r - 1.$$

More generally, a twist b replaces $\mathcal{O}_E(-1)$ by $\mathcal{O}_E(-1+b)$ and the intersection condition by $L_0 \cdot E = r + b - 1$; the case $b = 0$ is the untwisted case considered below.

The basic model is the contraction

$$\tau : \mathbb{F}_3 \longrightarrow \mathbb{P}(1, 1, 3),$$

which contracts the negative section $C_0 \subset \mathbb{F}_3$. In the notation of [Section 2](#), Karmazyn–Kuznetsov–Shinder give the full exceptional collection [[18](#), Example 3.17]

$$D^b(\mathrm{coh} \mathbb{F}_3) = \langle \mathcal{O}(-C_0 - f), \mathcal{O}(-f), \mathcal{O}, \mathcal{O}(C_0 + 3f) \rangle.$$

This collection differs from the projective-bundle collection of [Section 2](#). It is chosen so that $\mathcal{O}_{C_0}(-1)$ belongs to its first block, which is precisely the compatibility condition

needed for KKS descent; the same block will also satisfy the stronger condition of untwisted adherence. We group its objects into the three blocks

$$\tilde{\mathcal{A}}_0 = \langle \mathcal{O}(-C_0 - f), \mathcal{O}(-f) \rangle, \quad \tilde{\mathcal{A}}_1 = \langle \mathcal{O} \rangle, \quad \tilde{\mathcal{A}}_2 = \langle \mathcal{O}(C_0 + 3f) \rangle.$$

To check compatibility, we exhibit $\mathcal{O}_{C_0}(-1)$ as the quotient of a map between the two generators of $\tilde{\mathcal{A}}_0$. The standard exact sequence associated with the effective divisor C_0 , tensored by $\mathcal{O}(-f)$, is

$$0 \longrightarrow \mathcal{O}(-C_0 - f) \longrightarrow \mathcal{O}(-f) \longrightarrow \mathcal{O}_{C_0}(-f) \longrightarrow 0.$$

The restriction $\mathcal{O}(-f)|_{C_0}$ has degree $(-f) \cdot C_0 = -1$. Since $C_0 \simeq \mathbb{P}^1$, this identifies the quotient with $\mathcal{O}_{C_0}(-1)$:

$$\mathcal{O}_{C_0}(-f) \simeq \mathcal{O}_{C_0}(-1).$$

Since the first two terms of the exact sequence lie in $\tilde{\mathcal{A}}_0$, the associated distinguished triangle gives

$$\mathcal{O}_{C_0}(-1) \in \tilde{\mathcal{A}}_0.$$

The curve C_0 is the only irreducible curve contracted by τ . Consequently, this containment proves that the displayed decomposition is compatible with τ in the sense of [Definition 4.1](#).

We now verify the stronger untwisted-adherence condition. Since $\mathbb{F}_3$ is rational, $H^\bullet(\mathbb{F}_3, \mathcal{O}_{\mathbb{F}_3}) \cong \mathbb{C}$, and the negative section is a smooth rational curve with $C_0^2 = -3$. Set

$$L_0 = \mathcal{O}(-C_0 - f), \quad L_1 = \mathcal{O}(-f).$$

Here $L_1 = L_0(C_0)$, and the distinguished triangle associated with the exact sequence above yields

$$\tilde{\mathcal{A}}_0 = \langle L_0, L_1 \rangle = \langle L_0, \mathcal{O}_{C_0}(-1) \rangle.$$

The intersection conventions of [Section 2](#) also give

$$L_0 \cdot C_0 = (-C_0 - f) \cdot C_0 = 2 = 3 - 1.$$

The one-curve criterion stated above therefore shows directly that $\tilde{\mathcal{A}}_0$ is untwisted adherent to C_0 .

The compatibility established above allows us to apply [\[18, Theorem 2.12\]](#). The resulting decomposition is

$$D^b(\mathrm{coh} \mathbb{P}(1, 1, 3)) = \langle \mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2 \rangle,$$

where $\mathcal{A}_i$ is the component descended from $\tilde{\mathcal{A}}_i$. More precisely, the theorem describes each descended component as the Verdier quotient of $\tilde{\mathcal{A}}_i$ by the objects $\mathcal{O}_E(-1)$ associated with the exceptional curves assigned to that block. In the present decomposition, the curve C_0 is assigned to $\tilde{\mathcal{A}}_0$, whereas no exceptional curve is assigned to $\tilde{\mathcal{A}}_1$ or $\tilde{\mathcal{A}}_2$. Consequently, τ_* restricts to equivalences

$$\tilde{\mathcal{A}}_1 \xrightarrow{\sim} \mathcal{A}_1, \quad \tilde{\mathcal{A}}_2 \xrightarrow{\sim} \mathcal{A}_2.$$

Each of these two categories is generated by a single exceptional line bundle. Their descended components are therefore generated by exceptional objects and satisfy

$$\mathcal{A}_1 \simeq D^b(\mathbb{C}), \quad \mathcal{A}_2 \simeq D^b(\mathbb{C}).$$

Explicitly, these components are generated by $\mathcal{O}$ and $\mathcal{O}(3)$ on $\mathbb{P}(1, 1, 3)$, respectively, as in [18, Example 3.17]. The adherence of $\tilde{\mathcal{A}}_0$ supplies the additional information needed to identify the first component. The Kalck–Karmazyn algebra of the singularity $\frac{1}{3}(1, 1)$ is

$$K(3, 1) \cong \mathbb{C}[z_1, z_2]/(z_1, z_2)^2$$

by [18, Example 3.14]. The identification theorem [18, Theorem 3.16] therefore gives

$$\mathcal{A}_0 \simeq D^b(K(3, 1)\text{-mod}).$$

This is the decomposition described in [18, Example 3.17].

The next lemma shows that both the containment

$$\mathcal{O}_C(-1) \in \mathcal{A}$$

required for KKS compatibility and the untwisted-adherence condition are preserved by blow-ups whose centers are disjoint from C .

Lemma 4.3. *Let $\beta : S' \rightarrow S$ be the blow-up of a smooth point disjoint from a smooth rational curve $C \subset S$, and let $C' \subset S'$ be its strict transform. If*

$$\mathcal{O}_C(-1) \in \mathcal{A} \subset D^b(\text{coh } S),$$

then

$$\mathcal{O}_{C'}(-1) \in \beta^* \mathcal{A}.$$

Moreover, if $C^2 = -r$ and $\mathcal{A} = \langle L_0, L_1 \rangle$ is untwisted adherent to C , then $\beta^ \mathcal{A}$ is untwisted adherent to C' . The same conclusions hold after any finite sequence of blow-ups whose centers are disjoint from the successive strict transforms of C .*

Proof. Since β is an isomorphism in a neighborhood of C' , it induces identifications

$$C' \simeq C, \quad \beta^* \mathcal{O}_C(-1) \simeq \mathcal{O}_{C'}(-1).$$

It follows that $\mathcal{O}_{C'}(-1) \in \beta^* \mathcal{A}$. For untwisted adherence, the line-bundle data satisfy

$$L_1 = L_0(C), \quad L_0 \cdot C = r - 1.$$

Since $\beta^* C = C'$, their pullbacks satisfy

$$\beta^* L_1 = \beta^* L_0(C'), \quad (\beta^* L_0) \cdot C' = L_0 \cdot C = r - 1.$$

A point blow-up also preserves the cohomology of the structure sheaf. Consequently, the conditions of [18, Lemma 3.5 and Definition 3.6] remain valid on S' , and $\beta^* \mathcal{A}$ is untwisted adherent to C' . The final assertion follows by induction. $\square$

Remark 4.4 (Bright’s exact sequence). The Brauer-group criterion also gives the required vanishing for the one-point models used below. We use the cohomological Brauer group $\text{Br}(X) = H_{\text{ét}}^2(X, \mathbb{G}_m)_{\text{tors}}$. In the present projective setting, its torsion classes are represented by Azumaya algebras by Gabber’s theorem, as used in [4, proof of Theorem 6]; thus no distinction between the two Brauer groups arises for the finite groups computed here. This is also the Brauer group entering the obstruction in [18, Corollary 4.8].

We use Bright's exact sequence for a normal surface X with rational singularities. Let $\tau : Y \rightarrow X$ be the minimal resolution, and let $\Lambda_\tau \subset \text{Pic}(Y)$ denote the subgroup generated by the exceptional curves. The intersection pairing induces a homomorphism

$$\theta : \text{Pic}(Y) \longrightarrow \Lambda_\tau^\vee, \quad D \longmapsto (E \mapsto D \cdot E),$$

and Bright's exact sequence [4, Proposition 1] is

$$0 \longrightarrow \text{Pic}(X) \longrightarrow \text{Pic}(Y) \xrightarrow{\theta} \Lambda_\tau^\vee \longrightarrow \text{Br}(X) \longrightarrow \text{Br}(Y).$$

For the surfaces considered here, the same sequence holds with the full second cohomology groups: the group on Y vanishes and the cokernel of θ is finite, since the exceptional intersection matrix is nondegenerate. Thus taking torsion does not change the sequence. The vanishing $\text{Br}(Y) = 0$ for a smooth projective rational surface follows from birational invariance of the Brauer group and $\text{Br}(\mathbb{P}^2) = 0$; compare [4, Section 3].

Corollary 4.5. *Let X be a normal projective surface with quotient singularities and rational minimal resolution $\tau : Y \rightarrow X$. For the intersection map of Remark 4.4, there is an isomorphism*

$$\text{Br}(X) \cong \text{coker}(\theta : \text{Pic}(Y) \longrightarrow \Lambda_\tau^\vee).$$

In particular, $\text{Br}(X) = 0$ if and only if there are divisor classes $L_i \in \text{Pic}(Y)$ with $L_i \cdot E_j = \delta_{ij}$ for all exceptional curves E_j .

Proof. Quotient singularities are rational, so Bright's sequence applies, and $\text{Br}(Y) = 0$. Surjectivity of θ is equivalent to its image containing every element of the basis dual to the exceptional curves. $\square$

All log del Pezzo surfaces in this paper satisfy the rationality hypothesis by Proposition 2.2. In this setting, the vanishing in Corollary 4.5 is equivalent to the KKS *torsion-free* condition: the Weil divisor class group $\text{Cl}(X)$ has no torsion [18, Definition 4.7]. It is necessary for an untwisted adherent decomposition by [18, Corollary 4.8]; the absence of this obstruction alone does not construct such a decomposition. Replacing all L_i by $-L_i$ gives the equivalent convention $L_i \cdot E_j = -\delta_{ij}$, which is used in the explicit three-point calculation in Appendix A.3 and in the six-point obstruction in Appendix A.4.

For $X = \mathbb{P}(1, 1, 3)$, the minimal resolution is $\tau : \mathbb{F}_3 \rightarrow \mathbb{P}(1, 1, 3)$, and its exceptional locus is C_0 . The exceptional lattice is $\Lambda_\tau = \mathbb{Z}C_0$, and we identify $\Lambda_\tau^\vee$ with $\mathbb{Z}$ by evaluating on C_0 . The fiber class f satisfies

$$f \cdot C_0 = 1.$$

It follows that the map

$$\text{Pic}(\mathbb{F}_3) \longrightarrow \Lambda_\tau^\vee \cong \mathbb{Z}, \quad D \longmapsto D \cdot C_0,$$

is surjective. Since $\text{Br}(\mathbb{F}_3) = 0$, Bright's exact sequence gives

$$\text{Br}(\mathbb{P}(1, 1, 3)) = 0.$$

The corresponding calculation for X_{10} follows from its blow-up geometry in Section 5. The behavior of the obstruction in all six Corti–Heuberger cascades is analyzed in Appendix A.

5 Corti–Heuberger cascades and the surface X_{10}

This section proceeds in four steps. First, we recall the Corti–Heuberger classification and summarize the explicit rigid-base models in [Proposition 5.4](#), leaving their geometry to [Appendix B](#). Second, we identify X_{10} within the classification and establish its general Reid–Suzuki blow-up presentation in [Propositions 5.5 to 5.7](#). Third, we construct its full exceptional collection and compute the Euler form in [Corollary 5.8](#) and [Proposition 5.9](#). Finally, we verify compatibility and adherence, carry out KKS descent to the singular surface, and extend the decomposition along the one-point cascade in [Theorem 5.10](#) and [Corollary 5.11](#). Throughout, the construction follows the order already established: Orlov objects on the resolution, their images under Φ , and the preceding residual-gerbe objects on the canonical stack.

We first recall the deformation terminology underlying the classification.

Definition 5.1 (*qG-deformations and rigidity*). Let X be a normal projective surface with quotient singularities. A $\mathbb{Q}$ -Gorenstein deformation, or *qG-deformation*, of X is a flat deformation that locally lifts to equivariant deformations of the index-one covers of the quotient singularities. The index-one cover is the local cyclic cover on which the canonical divisor becomes Cartier. For projective surfaces we use proper families, and *qG-deformation equivalence* means being connected by a chain of such families over connected bases [[1](#), Basic Concepts, Definition 2 and proof of Lemma 6]. A surface is *qG-rigid* if every sufficiently small *qG-deformation* is trivial.

Rigidity of the surface is a global condition, distinct from rigidity of its singularities. The singularity $\frac{1}{3}(1, 1)$ is locally *qG-rigid*, so small *qG-deformations* preserve its analytic type; the degree is also locally constant [[7](#), Section 1.1.1].

Corti–Heuberger classify the non-smooth del Pezzo surfaces with only $\frac{1}{3}(1, 1)$ -points into 29 families, grouped into six cascades [[7](#), Theorems 3–4]. The operations relating these families are described by the following terminology.

Definition 5.2 (*Floating (-1) -curves and cascades*). Let X be a log del Pezzo surface with only $\frac{1}{3}(1, 1)$ -points. A *floating (-1) -curve* on X is a smooth rational curve contained in its smooth locus with self-intersection -1 [[7](#), Definition 5]. A *cascade* connects the deformation families by blow-ups at nonsingular points, or equivalently by reversing contractions of floating (-1) -curves.

Contracting a floating (-1) -curve is the inverse of a blow-up at a nonsingular point and increases K^2 by one. These operations preserve the singular points and their analytic types.

Definition 5.3 (*Marked blow-up presentations*). A *marked blow-up presentation* of a smooth surface Y is a sequence of point blow-ups

$$Y_0 \xleftarrow{\beta_1} Y_1 \xleftarrow{\beta_2} \cdots \xleftarrow{\beta_m} Y_m \simeq Y,$$

with fixed identifications of the starting and final surfaces and, at each stage, a specified point $q_i \in Y_{i-1}$, its incidences with the strict transforms of previously introduced exceptional curves, and a label for the new exceptional divisor $D_i \subset Y_i$. The centers may be infinitely near points of Y_0 . When Y is a resolution of a surface with quotient singularities, the marking also specifies the curves contracted to those singularities. A *marked cascade presentation* similarly specifies the successive smooth-point blow-ups and identifies its base with the chosen model of Z_k .

Thus a marking records actual geometric choices, not merely the starting surface, a number of blow-ups, or an abstract incidence pattern.

We follow the notation of [7, Theorem 4]: k denotes the number of $\frac{1}{3}(1, 1)$ points, and $d = K_X^2$ denotes the anticanonical degree of a surface X . In the Fano index one case, $X_{k,d}$ denotes the qG -deformation family with k singular points and degree d . The rigid bases are identified in [7, Theorem 6 and Remark 7], and [7, Corollary 8] organizes the remaining families into cascades of blow-ups from them. Each of these base families consists of a single isomorphism class, denoted by Z_k :

k	Z_k	$K_{Z_k}^2$
1	$S_{1,25/3} = \mathbb{P}(1, 1, 3)$	$25/3$
2	$X_{2,17/3}$	$17/3$
3	$X_{3,5}$	5
4	$X_{4,7/3}$	$7/3$
5	$X_{5,5/3}$	$5/3$
6	$X_{6,2}$	2.

Here the third column records the anticanonical degree of the rigid base. The notation $S_{1,25/3}$ distinguishes $\mathbb{P}(1, 1, 3)$, which has Fano index five, from the index-one families $X_{k,d}$.

Proposition 5.4. *Each minimal resolution $\tilde{Z}_k$ has an explicit marked presentation by successive point blow-ups of a smooth surface Y_0 , with infinitely near centers allowed, as follows:*

k	Z_k	Y_0	number of blow-ups	$\text{rk } K_0(\tilde{Z}_k)$
1	$\mathbb{P}(1, 1, 3)$	$\mathbb{F}_3$	0	4
2	$X_{2,17/3}$	$\mathbb{F}_3$	3	7
3	$X_{3,5}$	$\mathbb{F}_3$	4	8
4	$X_{4,7/3}$	$\mathbb{F}_2$	7	11
5	$X_{5,5/3}$	$\mathbb{P}^2$	9	12
6	$X_{6,2}$	$\mathbb{F}_0$	8	12

The marked incidence pattern of the centers is part of each presentation. The centers and the resulting Orlov full exceptional collections are given in Appendix B. Applying the Ishii–Ueda embedding and adjoining the residual-gerbe objects gives full exceptional collections on the canonical stacks of the bases, of lengths 5, 9, 11, 15, 17, 18, respectively.

The proof of Proposition 5.4, including the marked blow-up constructions and the explicit collections, is given in Appendix B.

The same starting surface and number of blow-ups need not determine the contracted configuration or the singularity basket. In particular, the three blow-ups of $\mathbb{F}_3$ used for Z_2 create a second (-3) -curve, unlike the marked configuration for a degree-16/3 member of the one-point cascade.

Every surface in an index-one family $X_{k,d}$ is obtained from its listed base by $K_{Z_k}^2 - d$ blow-ups at nonsingular points of the successive surfaces [7, Corollary 8]. The uniqueness of the bases is stronger than local rigidity of the singularities: it identifies any explicit model in a base family with Z_k . This will allow the computations in Appendix A to apply throughout each cascade. For the categorical construction, Proposition 5.4 fixes explicit initial collections. The remaining terms are the Orlov objects from the smooth blow-ups and the residual-gerbe objects over the stacky points; Corollary B.3 records the resulting order.

The two auxiliary families $B_{1,16/3}$ and $B_{2,8/3}$ have Fano index two and may also occur as endpoints of a directed MMP [7, Theorems 3 and 6]. By [7, Remark 9], a smooth-point blow-up of a member of either family admits an alternative sequence of four floating contractions to Z_1 , respectively Z_2 . Together with Z_1 , these are the three higher-index families in the classification.

For the inductive step, let X belong to the family $X_{k,d}$. It is obtained from Z_k by

$$K_{Z_k}^2 - d$$

blow-ups at nonsingular points. Passing to minimal resolutions gives a sequence of ordinary smooth blow-ups starting from $\tilde{Z}_k$. At each step, Orlov's blow-up formula adds the exceptional object associated with the new exceptional curve. If $\langle G_1, \dots, G_N \rangle$ is the resulting collection on the resolution, the Ishii–Ueda order is

$$\langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_k}, \Phi(G_1), \dots, \Phi(G_N) \rangle,$$

where $\mathcal{E}_{p_i} = \iota_{p_i*}(\mathcal{O}_{B\mu_3} \otimes \rho_2)$. This construction gives a full exceptional collection on the canonical stack of X , with length

$$\ell = 12 - K_X^2 + \frac{4k}{3}.$$

We specialize this construction to X_{10} . Let x_0, x_1, x_2, x_3 be the weighted coordinates on $\mathbb{P}(1, 2, 3, 5)$, with weights 1, 2, 3, 5.

Proposition 5.5. *A general hypersurface $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ of weighted degree 10 is quasismooth. Moreover, it has a unique singular point,*

$$p = [0 : 0 : 1 : 0],$$

and this singularity is of type $\frac{1}{3}(1, 1)$.

Proof. We use the standard quasismoothness criterion for weighted hypersurfaces as in [15]. The base locus of $|\mathcal{O}_{\mathbb{P}(1,2,3,5)}(10)|$ is the single point $p = [0 : 0 : 1 : 0]$. On the smooth locus of the ambient weighted projective space away from p , Bertini's theorem therefore gives smoothness of a general member.

Since $\mathbb{P}(1, 2, 3, 5)$ is well formed and its weights are pairwise coprime, its singular locus consists of the three coordinate points of weights 2, 3, 5. The monomials x_1^5 and x_3^2 show that a general hypersurface does not pass through $[0 : 1 : 0 : 0]$ or $[0 : 0 : 0 : 1]$. It remains to analyze the point p . On the chart $x_2 \neq 0$, the local quotient cover is $\mathbb{A}^3$ with coordinates corresponding to x_0, x_1, x_3 , and the stabilizer μ_3 acts with weights

$$(1, 2, 5) \equiv (1, 2, 2) \pmod{3}.$$

The monomial $x_0 x_2^3$ has the required weighted degree. Its coefficient is therefore nonzero in a general defining polynomial. It gives a nonzero linear term in the coordinate corresponding to x_0 , so the hypersurface is smooth on the local quotient cover and we may eliminate this coordinate. The induced μ_3 -action on the tangent coordinates has weights (2, 2), which is equivalent to (1, 1). Thus X_{10} is quasismooth and its unique singularity is of type $\frac{1}{3}(1, 1)$. $\square$

By [Proposition 5.5](#), a general degree-10 hypersurface $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ is quasismooth and has a unique singularity of type $\frac{1}{3}(1, 1)$. Weighted adjunction and the intersection formula of [\[10\]](#) give

$$\omega_{X_{10}}^\vee = \mathcal{O}_{X_{10}}(1), \quad K_{X_{10}}^2 = \frac{10}{1 \cdot 2 \cdot 3 \cdot 5} = \frac{1}{3}.$$

The anticanonical class is ample, so X_{10} is a log del Pezzo surface. The higher-index families in [\[7, Theorem 3\]](#) have different numerical invariants, so the Fano index is one; [\[7, Theorem 4 and Table 2\]](#) identifies the family as $X_{1,1/3}$.

The family $X_{1,1/3}$ is the last index-one family of the one-point cascade. By [\[7, Corollary 8\(1\)\]](#), its members are obtained from $\mathbb{P}(1, 1, 3)$ by $25/3 - 1/3 = 8$ blow-ups of nonsingular points. This presentation does not by itself establish that a general hypersurface comes from general centers. [Proposition 5.6](#) constructs the hypersurface from eight general points, whereas [Proposition 5.7](#) establishes the converse for a general hypersurface by a moduli-dimension and dominance argument; this converse is not a formal consequence of the cascade classification. Together they supply the distinctness needed for the Euler-form computation in [Proposition 5.9](#) and the disjoint curves used in [Theorem 5.10](#).

A chosen directed MMP in the one-point case may instead terminate at the auxiliary family $B_{1,16/3}$. This does not conflict with the presentation above. Indeed, [\[7, Remark 9\(1\)\]](#) states that a blow-up $X \rightarrow B_{1,16/3}$ of a nonsingular point admits an alternative sequence of four contractions of floating (-1) -curves from X to $\mathbb{P}(1, 1, 3)$. The available (-1) -curves therefore permit different contraction sequences, so the endpoint of one directed MMP does not obstruct the eight-blow-up presentation.

Proposition 5.6 (The Reid–Suzuki construction). *For eight distinct general smooth points $P_1, \dots, P_8 \in \mathbb{P}(1, 1, 3)$, their blow-up $\eta : S \rightarrow \mathbb{P}(1, 1, 3)$ is a log del Pezzo surface in $X_{1,1/3}$. Its anticanonical divisor and ring are*

$$-K_S = \eta^*(-K_{\mathbb{P}(1,1,3)}) - \sum_{i=1}^8 D_i,$$

$$R(S, -K_S) \cong \frac{\mathbb{C}[x, y, z, w]}{(f_{10})}, \quad \deg(x, y, z, w) = (1, 2, 3, 5),$$

where D_i is the exceptional curve over P_i and f_{10} is weighted homogeneous of degree 10. The surface S is isomorphic to its anticanonical model, a quasismooth hypersurface $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$.

Proof. We first justify ampleness independently of the ring presentation. By the preceding identification of the hypersurface family and [\[7, Corollary 8\(1\)\]](#), some member of $X_{1,1/3}$ is obtained by eight smooth-point blow-ups of $\mathbb{P}(1, 1, 3)$. These centers must be distinct on the original surface: a blow-up on an earlier exceptional curve would leave its strict transform with anticanonical degree at most zero, contradicting ampleness on the final surface. The space of ordered distinct smooth eight-tuples is irreducible, and ampleness of the Cartier divisor $-3K$ is open in the corresponding family of blow-ups [\[24, Tag 0D2S\]](#). Hence $-K_S$ is ample for general centers. The blow-up formula gives the asserted divisor and

$$K_S^2 = \frac{25}{3} - 8 = \frac{1}{3}.$$

The blow-ups preserve the unique $\frac{1}{3}(1,1)$ -point and introduce no other singularities. Moreover, $(-K_S) \cdot D_i = 1$ forces Fano index one, so S belongs to $X_{1,1/3}$.

The anticanonical-ring presentation for general centers is the Reid–Suzuki construction [23, Section 1 and Table 1.1]. Since $-K_S$ is ample, its Proj is S itself. The resulting hypersurface is quasismooth: away from the vertex, the anticanonical cone is locally a $\mathbb{G}_m$ -bundle over the smooth locus. Over the $\frac{1}{3}(1,1)$ -point it is locally the quotient of $\mathbb{A}^2 \times \mathbb{G}_m$ by μ_3 , acting faithfully on the second factor through the canonical character. This action is free, so the punctured cone is smooth there as well. $\square$

The following proposition gives the converse of Proposition 5.6 for a general quasismooth hypersurface.

Proposition 5.7 (The general hypersurface). *A general quasismooth $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ is isomorphic to the anticanonical model of the blow-up of $\mathbb{P}(1, 1, 3)$ at eight distinct general smooth points. Its minimal resolution is therefore the blow-up of $\mathbb{F}_3$ at eight distinct points away from the negative section C_0 .*

Proof. Let U be the nonempty open set of quasismooth degree-10 forms in $\mathbb{P}H^0(\mathbb{P}(1, 2, 3, 5), \mathcal{O}(10))$. In coordinates x, y, z, w of weights 1, 2, 3, 5, a degree-10 monomial $x^a y^b z^c w^e$ satisfies $a + 2b + 3c + 5e = 10$. For fixed e, c , the number of choices of b , with a then determined, is $\lfloor (10 - 5e - 3c)/2 \rfloor + 1$. Thus the contributions for $e = 0, 1, 2$ are respectively

$$(6 + 4 + 3 + 1), \quad (3 + 2), \quad 1,$$

giving $14 + 5 + 1 = 20$ monomials. Hence U , being a nonempty open subset of $\mathbb{P}^{19}$, is irreducible of dimension 19. The group $G = \text{Aut } \mathbb{P}(1, 2, 3, 5)$ is the group of graded coordinate changes modulo weighted scalar multiplication [8, Theorem 4.2], and has dimension

$$\dim G = (1 + 2 + 3 + 6) - 1 = 11.$$

The summands count forms of degrees 1, 2, 3, 5, respectively. Isomorphic hypersurfaces in U lie in the same G -orbit: an isomorphism preserves the anticanonical class and hence its section ring, and a graded isomorphism of these rings lifts to a graded polynomial automorphism because their generators have degree less than the relation degree 10.

Let W be a nonempty invariant open subset of the space of ordered distinct smooth eight-tuples on $\mathbb{P}(1, 1, 3)$ for which Proposition 5.6 holds, and write S_P for the blow-up associated with $P \in W$. We have $\dim W = 16$ and

$$\dim H = 8, \quad H = \text{Aut } \mathbb{P}(1, 1, 3),$$

since its graded automorphisms have the form $(u, v) \mapsto A(u, v)$, $z \mapsto az + q_3(u, v)$, modulo weighted scalars, again by [8, Theorem 4.2]. Here $A \in \text{GL}_2$, $a \neq 0$, and q_3 is a binary cubic. The stabilizer of a general ordered tuple is trivial. Indeed, three distinct images in $\mathbb{P}_{[u:v]}^1$ force the induced projective transformation to be the identity. On the chart $u \neq 0$, the remaining action is

$$(t, \zeta) \mapsto (t, a\zeta + q(t)), \quad \deg q \leq 3.$$

Writing $q(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3$, the fixing equations are

$$(a - 1)\zeta_j + c_0 + c_1 t_j + c_2 t_j^2 + c_3 t_j^3 = 0, \quad 1 \leq j \leq 8.$$

The coefficient matrix has rows $(\zeta_j, 1, t_j, t_j^2, t_j^3)$. For five distinct t_j with $\zeta_j = t_j^4$, the corresponding minor, after a column permutation, is the Vandermonde determinant

$$\det(t_j^{\ell-1})_{1 \leq j, \ell \leq 5} = \prod_{1 \leq i < j \leq 5} (t_j - t_i) \neq 0.$$

Its nonvanishing therefore defines a nonempty open subset of the irreducible configuration space of ordered distinct smooth eight-tuples, which meets the nonempty open set W . Thus the matrix has rank five for a general tuple in W , forcing $a = 1$ and $q = 0$.

Each S_P has only finitely many floating (-1) -curves. Indeed, pull back $-K_{S_P}$ to its minimal resolution and denote its class by A_P . It has square $1/3$, and the classes of floating curves satisfy $C^2 = -1$ and $A_P \cdot C = 1$. The Hodge index theorem [13, Chapter V, Theorem 1.9] makes $A_P^\perp$ negative definite in the numerical divisor space, so these conditions allow only finitely many integral classes; a negative class contains at most one irreducible curve. Consequently, each isomorphism class of S_P corresponds to finitely many H -orbits of tuples, obtained by choosing an ordered set of eight disjoint floating curves to contract. Rosenlicht's theorem gives a geometric quotient on a dense invariant open subset of W , of dimension $16 - 8 = 8$ [5, Section 1.2]. The finite ambiguity in the choice of contracted curves does not change this dimension, so the isomorphism classes also vary in dimension eight. Moreover, a finite-index subgroup of $\text{Aut}(S_P)$ preserves each of the eight chosen curves and descends to the stabilizer of P in H ; hence $\text{Aut}(S_P)$ is finite for general P .

Let $V \subset U$ consist of the forms whose hypersurfaces are isomorphic to some S_P , with $P \in W$. After shrinking W , choosing generators of the relative anticanonical rings locally and applying graded coordinate changes gives an algebraic family of equations with image V . Chevalley's theorem [24, Tag 054K] shows that V is constructible, and it is G -stable by construction. Since the general stabilizer is finite, the orbit-dimension formula [5, Proposition 1.11] gives general orbit dimension 11. Applying the same geometric-quotient dimension formula to this family, whose isomorphism classes vary in dimension eight, gives

$$\dim V = 8 + 11 = 19 = \dim U.$$

Thus V is dense and constructible in U , so it contains a dense open subset [24, Tag 005K], proving the converse.

The minimal resolution $\tau : \mathbb{F}_3 \rightarrow \mathbb{P}(1, 1, 3)$ is an isomorphism over the smooth locus. Since all eight centers are distinct and lie in that locus, they lift uniquely to eight distinct points $q_1, \dots, q_8 \in \mathbb{F}_3$ away from the negative section C_0 . Hence the minimal resolution is the blow-up of $\mathbb{F}_3$ at these eight points. $\square$

Let

$$\sigma : \tilde{X}_{10} \longrightarrow \mathbb{F}_3$$

be the blow-down morphism supplied by Proposition 5.7 for a general X_{10} , let $D_1, \dots, D_8$ be the pairwise disjoint exceptional curves of its eight point blow-ups, and let f be the fiber class on $\mathbb{F}_3$. Let

$$\nu : \tilde{X}_{10} \longrightarrow X_{10}$$

be the minimal resolution, and let E be the strict transform of C_0 . The geometric description also gives

$$\text{Br}(X_{10}) = 0.$$

Indeed, E is the only curve contracted by ν , and also we have

$$(\sigma^* f) \cdot E = f \cdot C_0 = 1.$$

This class makes the intersection map in [Remark 4.4](#) surjective. Since $\tilde{X}_{10}$ is rational, Bright's exact sequence gives the claimed vanishing.

Pulling back the standard full exceptional collection

$$\langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + 3f), \mathcal{O}(C_0 + 4f) \rangle$$

and applying Orlov's blow-up formula gives the collection

$$\langle \mathcal{O}_{D_1}(-1), \dots, \mathcal{O}_{D_8}(-1), \sigma^* \mathcal{O}, \sigma^* \mathcal{O}(f), \sigma^* \mathcal{O}(C_0 + 3f), \sigma^* \mathcal{O}(C_0 + 4f) \rangle$$

on $\tilde{X}_{10}$. The curves D_i are pairwise disjoint, so their exceptional objects may be listed in any order. Therefore, [Theorem 1.1](#) places the local object $\mathcal{E}_p$ before the image of this collection and gives the following consequence.

Corollary 5.8. *Let $\pi : \mathcal{X}_{10} \rightarrow X_{10}$ be the canonical stack. The category $D^b(\text{coh } \mathcal{X}_{10})$ has a full exceptional collection of length 13.*

More generally, the Hille–Perling toric systems on $\mathbb{F}_3$ induce the following family of full exceptional collections on $\tilde{X}_{10}$ [[14](#), Proposition 5.2]:

$$\langle \mathcal{O}_{D_1}(-1), \dots, \mathcal{O}_{D_8}(-1), \sigma^* \mathcal{O}, \sigma^* \mathcal{O}(f), \sigma^* \mathcal{O}(C_0 + (s+4)f), \sigma^* \mathcal{O}(C_0 + (s+5)f) \rangle,$$

for all $s \in \mathbb{Z}$.

Proposition 5.9 (The Euler form for $\mathcal{X}_{10}$). *Let $s \in \mathbb{Z}$ and let*

$$\mathcal{C}(s) = \langle \mathcal{E}_p, \Phi(\mathcal{O}_{D_1}(-1)), \dots, \Phi(\mathcal{O}_{D_8}(-1)), \Phi(\sigma^* \mathcal{O}), \Phi(\sigma^* \mathcal{O}(f)), \Phi(\sigma^* \mathcal{O}(C_0 + (s+4)f)), \Phi(\sigma^* \mathcal{O}(C_0 + (s+5)f)) \rangle$$

be an exceptional collection of $D^b(\text{coh } \mathcal{X}_{10})$ associated with the Hille–Perling toric system for $\mathbb{F}_3$. The matrix of the Euler form $S_{ij}(s) = \chi(\mathcal{C}_i(s), \mathcal{C}_j(s))$ is given by

$$S_{\mathcal{X}_{10}}(s) = \left(\begin{array}{c|c|cccc} 1 & 0_{1 \times 8} & 0 & -1 & -(s+1) & -(s+2) \\ \hline 0_{8 \times 1} & I_8 & & & & -\mathbf{1}_{8 \times 4} \\ \hline & & 1 & 2 & 2s+7 & 2s+9 \\ & & 0 & 1 & 2s+5 & 2s+7 \\ & & 0 & 0 & 1 & 2 \\ & & 0 & 0 & 0 & 1 \end{array} \right).$$

Equivalently, with respect to the ordered collection $\mathcal{C}(s)$, the fully expanded matrix is

$$S_{\mathcal{X}_{10}}(s) = \left(\begin{array}{c|cccccccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -(s+1) & -(s+2) \\ \hline 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & -1 & -1 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 2s+7 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2s+5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right).$$

Proof. We compute the matrix block by block: first the row and column involving the residual-gerbe object, then the diagonal block of the eight blow-up objects, the pairings between those objects and the four pulled-back line bundles, and finally the 4×4 Hirzebruch-surface block.

Let

$$\Psi : D^b(\text{coh } \mathcal{X}_{10}) \longrightarrow D^b(\text{coh } \tilde{\mathcal{X}}_{10})$$

be the left adjoint of the Ishii–Ueda functor

$$\Phi : D^b(\text{coh } \tilde{\mathcal{X}}_{10}) \longrightarrow D^b(\text{coh } \mathcal{X}_{10}).$$

We first compute the entries involving the residual-gerbe object $\mathcal{E}_p$. Locally at p , the stack is $[\mathbb{A}^2/\mu_3]$, and $\mathcal{E}_p$ is represented by $\mathcal{O}_0 \otimes \rho_2$, where ρ_2 is the unique non-special representation. The left-adjoint computation of Gugiatti–Rota [12, Theorem 3.1 and Example 3.5] applies. For the singularity $\frac{1}{3}(1, 1)$, the natural and determinant representations are

$$\rho_{\text{nat}} = \rho_1 \oplus \rho_1, \quad \rho_{\text{det}} = \bigwedge^2 \rho_{\text{nat}} = \rho_2.$$

The minimal resolution has a single exceptional curve E , with $E^2 = -3$, and the special representations are ρ_0 and ρ_1 . The relation

$$\rho_2 \otimes \rho_{\text{det}} = \rho_2 \otimes \rho_2 = \rho_1$$

gives

$$\Psi(\mathcal{E}_p) \simeq \mathcal{O}_E(-2).$$

Equivalently, [12, Example 3.5] states directly in its local notation that, for a cyclic quotient singularity of type $\frac{1}{n}(1, 1)$, we have

$$\psi(\mathcal{O}_0 \otimes \rho_{n-1}) \simeq \mathcal{O}_E(-n+1).$$

Taking $n = 3$ gives the displayed identification $\Psi(\mathcal{E}_p) \simeq \mathcal{O}_E(-2)$, with no cohomological shift. Thus, for every $A \in D^b(\text{coh } \tilde{\mathcal{X}}_{10})$, adjunction gives

$$\chi_{\mathcal{X}_{10}}(\mathcal{E}_p, \Phi A) = \chi_{\tilde{\mathcal{X}}_{10}}(\mathcal{O}_E(-2), A).$$

On the other hand, the Ishii–Ueda semiorthogonal decomposition has the form

$$D^b(\mathrm{coh} \mathcal{X}_{10}) = \left\langle \mathcal{E}_p, \Phi(D^b(\mathrm{coh} \tilde{X}_{10})) \right\rangle.$$

In the ordering of [16, Theorem 1.4], the exceptional complement lies to the left of $\Phi(D^b(\mathrm{coh} \tilde{X}_{10}))$. Consequently, semiorthogonality gives

$$\chi_{\mathcal{X}_{10}}(\Phi A, \mathcal{E}_p) = 0.$$

It remains to compute only the first row.

The curve E is the strict transform of the negative section $C_0 \subset \mathbb{F}_3$, while the curves $D_1, \dots, D_8$ are disjoint from E . Hence the corresponding Euler characteristics vanish:

$$\chi_{\tilde{X}_{10}}(\mathcal{O}_E(-2), \mathcal{O}_{D_i}(-1)) = 0$$

for all i . Let $L = \mathcal{O}_{\mathbb{F}_3}(D)$ be a line bundle on $\mathbb{F}_3$. Since E is identified with C_0 under σ , its restricted degree is

$$\deg((\sigma^*L)|_E) = D \cdot C_0.$$

For the inclusion $i : E \hookrightarrow \tilde{X}_{10}$, Grothendieck duality for an effective Cartier divisor gives

$$i^!(\sigma^*L) \simeq (\sigma^*L)|_E \otimes \mathcal{O}_E(E)[-1].$$

Since $\mathcal{O}_E(E) \simeq \mathcal{O}_E(-3)$, this gives

$$\mathrm{RHom}_{\tilde{X}_{10}}(\mathcal{O}_E(-2), \sigma^*L) \simeq \mathrm{R}\Gamma(E, \mathcal{O}_E(D \cdot C_0 - 1))[-1].$$

Consequently, the Euler characteristic is

$$\chi_{\tilde{X}_{10}}(\mathcal{O}_E(-2), \sigma^*L) = -\chi(\mathcal{O}_{\mathbb{P}^1}(D \cdot C_0 - 1)) = -D \cdot C_0.$$

Consider the following four line bundles:

$$\mathcal{O}, \quad \mathcal{O}(f), \quad \mathcal{O}(C_0 + (s+4)f), \quad \mathcal{O}(C_0 + (s+5)f).$$

Their intersection numbers with C_0 are

$$0, \quad 1, \quad s+1, \quad s+2.$$

These four values give the first row of the matrix.

For the block coming from the resolution, full faithfulness of Φ gives

$$\chi_{\mathcal{X}_{10}}(\Phi A, \Phi B) = \chi_{\tilde{X}_{10}}(A, B)$$

for all $A, B \in D^b(\mathrm{coh} \tilde{X}_{10})$. Since the D_i are the pairwise disjoint exceptional curves of the eight point blow-ups, their Euler pairings satisfy

$$\chi(\mathcal{O}_{D_i}(-1), \mathcal{O}_{D_j}(-1)) = \delta_{ij}.$$

For a line bundle L on $\mathbb{F}_3$, the restriction of its pullback satisfies $(\sigma^*L)|_{D_i} \simeq \mathcal{O}_{D_i}$. Using $D_i^2 = -1$, the same divisor-duality computation gives

$$\mathrm{RHom}_{\tilde{X}_{10}}(\mathcal{O}_{D_i}(-1), \sigma^*L) \simeq \mathrm{R}\Gamma(D_i, \mathcal{O}_{D_i})[-1],$$

and hence we obtain

$$\chi(\mathcal{O}_{D_i}(-1), \sigma^*L) = -1.$$

The opposite derived Hom is

$$\mathrm{RHom}_{\tilde{X}_{10}}(\sigma^*L, \mathcal{O}_{D_i}(-1)) \simeq \mathrm{R}\Gamma(D_i, \mathcal{O}_{D_i}(-1)),$$

and therefore we get

$$\chi(\sigma^*L, \mathcal{O}_{D_i}(-1)) = 0.$$

This gives the blocks I_8 , $-\mathbf{1}_{8 \times 4}$, and $0_{4 \times 8}$.

It remains to compute the 4×4 block coming from $\mathbb{F}_3$. The intersection data are

$$C_0^2 = -3, \quad C_0 \cdot f = 1, \quad f^2 = 0, \quad K_{\mathbb{F}_3} = -2C_0 - 5f.$$

For line bundles $L_1 = \mathcal{O}(A_1)$ and $L_2 = \mathcal{O}(A_2)$ on $\mathbb{F}_3$, Riemann–Roch gives

$$\chi(L_1, L_2) = \chi(\mathcal{O}(A_2 - A_1)) = 1 + \frac{1}{2}(A_2 - A_1) \cdot (A_2 - A_1 - K_{\mathbb{F}_3}).$$

Applying this formula to

$$\mathcal{O}, \quad \mathcal{O}(f), \quad \mathcal{O}(C_0 + (s+4)f), \quad \mathcal{O}(C_0 + (s+5)f),$$

gives

$$\begin{pmatrix} 1 & 2 & 2s+7 & 2s+9 \\ 0 & 1 & 2s+5 & 2s+7 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Combining the four blocks proves the stated formula. $\square$

We finally apply the KKS descent construction of [Section 4](#) to the singular surface X_{10} . The containment and adherence statements needed after the eight blow-ups are supplied by [Lemma 4.3](#).

Theorem 5.10. *The singular surface X_{10} admits a semiorthogonal decomposition*

$$D^b(\mathrm{coh} X_{10}) = \left\langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{10} \right\rangle,$$

where $F_1, \dots, F_{10}$ are exceptional objects.

Proof. Consider the KKS-adapted semiorthogonal decomposition

$$D^b(\mathrm{coh} \mathbb{F}_3) = \left\langle \tilde{\mathcal{A}}_0, \tilde{\mathcal{A}}_1, \tilde{\mathcal{A}}_2 \right\rangle,$$

introduced in [Section 4](#). By [Proposition 5.7](#), the morphism $\sigma : \tilde{X}_{10} \rightarrow \mathbb{F}_3$ is a sequence of eight blow-ups whose centers are disjoint from C_0 and its successive strict transforms. Applying the first part of [Lemma 4.3](#) repeatedly places $\mathcal{O}_E(-1)$ in $\sigma^*\tilde{\mathcal{A}}_0$, where E is the strict transform of C_0 . Separately, the second part of the lemma shows that $\sigma^*\tilde{\mathcal{A}}_0$ is untwisted adherent to E .

For a point blow-up $\beta : S' \rightarrow S$ with exceptional curve D , we use the right-handed form of Orlov’s blow-up formula

$$D^b(\mathrm{coh} S') = \left\langle \beta^*D^b(\mathrm{coh} S), \mathcal{O}_D \right\rangle.$$

This is equivalently obtained by right-mutating the Orlov component $\langle \mathcal{O}_D(-1) \rangle$ through $\beta^* D^b(\text{coh } S)$ [20, Theorem 1.6]. Since the eight blow-up centers are pairwise distinct, the resulting exceptional curves are pairwise disjoint, and the corresponding Orlov components may be listed in the order $D_1, \dots, D_8$. Iterating the right-handed decomposition therefore gives

$$D^b(\text{coh } \tilde{X}_{10}) = \langle \sigma^* \tilde{\mathcal{A}}_0, \sigma^* \tilde{\mathcal{A}}_1, \sigma^* \tilde{\mathcal{A}}_2, \mathcal{O}_{D_1}, \dots, \mathcal{O}_{D_8} \rangle.$$

The curves $D_1, \dots, D_8$ are the exceptional curves of the point blow-ups defining σ , but they are not contracted by $\nu : \tilde{X}_{10} \rightarrow X_{10}$. The only irreducible curve contracted by ν is E , and the preceding application of Lemma 4.3 gives

$$\mathcal{O}_E(-1) \in \sigma^* \tilde{\mathcal{A}}_0.$$

These two facts are precisely the compatibility condition of [18, Definition 2.7]. Hence the displayed semiorthogonal decomposition is compatible with the contraction ν .

The descent theorem [18, Theorem 2.12] therefore produces a semiorthogonal decomposition of $D^b(\text{coh } X_{10})$. The untwisted adherent component descending from $\sigma^* \tilde{\mathcal{A}}_0$ is equivalent to $D^b(K(3, 1)\text{-mod})$ by [18, Theorem 3.16].

The two components descending from $\sigma^* \tilde{\mathcal{A}}_1$ and $\sigma^* \tilde{\mathcal{A}}_2$, followed by the eight Orlov components, are exceptional. Thus there are ten exceptional components in total: two from the singleton KKS blocks and eight generated by $\mathcal{O}_{D_1}, \dots, \mathcal{O}_{D_8}$. Relabeling them in this order as $F_1, \dots, F_{10}$ gives the stated decomposition. $\square$

The preceding argument applies uniformly along the one-point cascade: the untwisted adherent component associated with the unique $\frac{1}{3}(1, 1)$ -singularity is unchanged, while each floating blow-up contributes one additional exceptional component.

Corollary 5.11. *Let X be a log del Pezzo surface belonging to a Corti–Heuberger cascade with exactly one singularity of type $\frac{1}{3}(1, 1)$. Suppose that the cascade from X to $\mathbb{P}(1, 1, 3)$ contains m contractions of floating (-1) -curves. Then*

$$D^b(\text{coh } X) = \langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{m+2} \rangle,$$

where $F_1, \dots, F_{m+2}$ are exceptional objects.

6 Further consequences

This section gives independent extensions of the preceding results and is not needed for the proofs of Theorems 1.1 to 1.4. The Ishii–Ueda decomposition and the rationality of the minimal resolution have the following consequence. For singularities of type $\frac{1}{k}(1, 1)$, both the local contribution and the length can be written explicitly.

Proposition 6.1. *Let X be a log del Pezzo surface, and let $\mathcal{X}$ be its canonical stack. The category $D^b(\text{coh } \mathcal{X})$ admits a full exceptional collection.*

Assume moreover that the singularities of X are $p_1, \dots, p_r$, where p_i has type $\frac{1}{k_i}(1, 1)$, with $k_i \geq 2$. The length of this full exceptional collection is

$$12 - K_X^2 + \sum_{i=1}^r \frac{2(k_i - 1)(k_i - 2)}{k_i}.$$

Equivalently, for the minimal resolution $f : \tilde{X} \rightarrow X$, we have

$$\mathrm{rk} K_0(\tilde{X}) = 12 - K_X^2 + \sum_{i=1}^r \frac{(k_i - 2)^2}{k_i},$$

and the canonical stack contributes $k_i - 2$ further exceptional objects over the point p_i .

Proof. By Proposition 2.2 and Corollary 2.6, the minimal resolution is a smooth rational surface with a full exceptional collection. The first assertion therefore follows from Theorem 3.4.

Under the additional hypothesis, let $E_i \subset \tilde{X}$ be the exceptional curve over p_i . Thus $E_i^2 = -k_i$, and the discrepancy formula is

$$K_{\tilde{X}} = f^* K_X - \sum_{i=1}^r \frac{k_i - 2}{k_i} E_i.$$

To verify the coefficient, write it as a_i and intersect with E_i . Adjunction gives

$$K_{\tilde{X}} \cdot E_i = -2 - E_i^2 = k_i - 2,$$

whereas $f^* K_X \cdot E_i = 0$. Hence $-k_i a_i = k_i - 2$, which gives $a_i = -\frac{k_i - 2}{k_i}$. The curves E_i are disjoint, so it follows that

$$K_{\tilde{X}}^2 = K_X^2 - \sum_{i=1}^r \frac{(k_i - 2)^2}{k_i}.$$

Consequently, the rank is

$$\mathrm{rk} K_0(\tilde{X}) = 12 - K_{\tilde{X}}^2 = 12 - K_X^2 + \sum_{i=1}^r \frac{(k_i - 2)^2}{k_i}.$$

For a singularity of type $\frac{1}{k_i}(1, 1)$, Gugiatti–Rota [12, Example 2.6] identify ρ_0 and ρ_1 as the special representations and order the local exceptional collection as

$$\langle \mathcal{O}_0 \otimes \rho_2, \dots, \mathcal{O}_0 \otimes \rho_{k_i-1} \rangle.$$

In the Ishii–Ueda decomposition, this local block precedes $\Phi(D^b(\mathrm{coh} \tilde{X}))$, and blocks at distinct singular points are mutually orthogonal. Hence p_i contributes $k_i - 2$ objects. Adding these local contributions to the rank above gives

$$12 - K_X^2 + \sum_{i=1}^r \left(\frac{(k_i - 2)^2}{k_i} + k_i - 2 \right) = 12 - K_X^2 + \sum_{i=1}^r \frac{2(k_i - 1)(k_i - 2)}{k_i}. \quad \square$$

The preceding proposition determines the length of the collection. In the one-point case, when the minimal resolution is obtained from $\mathbb{F}_n$ by blowing up pairwise distinct points away from the negative section, the same construction also gives a closed formula for its Euler matrix.

Proposition 6.2 (An Euler matrix for one cyclic point). *Let $n \geq 3$ and $m \geq 0$. Suppose that the minimal resolution of a log del Pezzo surface X fits into a diagram*

$$\tilde{X} \xrightarrow{\sigma} \mathbb{F}_n, \quad \tilde{X} \xrightarrow{\nu} X,$$

where σ is the blow-up of m distinct points of $\mathbb{F}_n$, all disjoint from the negative section $C_0 \subset \mathbb{F}_n$, and ν contracts the strict transform E of C_0 to the unique singular point of X , of type $\frac{1}{n}(1,1)$. Write $D_1, \dots, D_m$ for the exceptional curves of σ , and use the intersection conventions

$$C_0^2 = -n, \quad C_0 \cdot f = 1, \quad f^2 = 0.$$

Let $\mathcal{X}$ be the canonical stack of X , let $\Phi : D^b(\text{coh } \tilde{X}) \rightarrow D^b(\text{coh } \mathcal{X})$ be the Ishii–Ueda functor, and let $\iota : B\mu_n \hookrightarrow \mathcal{X}$ be the residual gerbe. For $2 \leq i \leq n-1$, define

$$\mathcal{E}_i := \iota_*(\mathcal{O}_{B\mu_n} \otimes \rho_i), \quad \rho_i(\zeta) = \zeta^i.$$

For $s \in \mathbb{Z}$, the Hille–Perling collection [14, Proposition 5.2], Orlov’s blow-up formula, and the Ishii–Ueda decomposition give the ordered full exceptional collection

$$\begin{aligned} \mathcal{C}_{n,m}(s) = \langle & \mathcal{E}_2, \dots, \mathcal{E}_{n-1}, \Phi(\mathcal{O}_{D_1}(-1)), \dots, \Phi(\mathcal{O}_{D_m}(-1)), \\ & \Phi(\sigma^*\mathcal{O}), \Phi(\sigma^*\mathcal{O}(f)), \Phi(\sigma^*\mathcal{O}(C_0 + (s+n+1)f)), \\ & \Phi(\sigma^*\mathcal{O}(C_0 + (s+n+2)f)) \rangle. \end{aligned}$$

The local block $T_n \in M_{n-2}(\mathbb{Z})$ is defined by

$$T_n := (\chi(\mathcal{E}_i, \mathcal{E}_j))_{2 \leq i, j \leq n-1}$$

and has entries

$$(T_n)_{ij} = \delta_{ij} - 2\delta_{j,i+1} + \delta_{j,i+2}.$$

Here the Kronecker symbols refer to the displayed integer range rather than to indices modulo n . Equivalently, with the evident truncation in the small dimensions,

$$T_n = \begin{pmatrix} 1 & -2 & 1 & 0 & \cdots & 0 \\ 0 & 1 & -2 & 1 & \ddots & \vdots \\ 0 & 0 & 1 & -2 & \ddots & 0 \\ \vdots & & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & 1 & -2 & \\ 0 & \cdots & \cdots & 0 & 1 & \end{pmatrix}. \quad (1)$$

For $n \geq 4$, the local-to-Hirzebruch block $B_n(s) \in \text{Mat}_{(n-2) \times 4}(\mathbb{Z})$ is

$$B_n(s) = \begin{pmatrix} & & 0_{(n-4) \times 4} & \\ 1 & 2 & s+2 & s+3 \\ 0 & -1 & -(s+1) & -(s+2) \end{pmatrix}, \quad (2)$$

where the zero block is absent for $n = 4$. The penultimate and final rows are indexed by $\mathcal{E}_{n-2}$ and $\mathcal{E}_{n-1}$, respectively. For $n = 3$, the local block consists only of $\mathcal{E}_2$, so $T_3 = (1)$. The representation $\rho_{n-2} = \rho_1$ is special and does not belong to the Ishii–Ueda complement; consequently, only the final row of $B_n(s)$ occurs:

$$B_3(s) = \begin{pmatrix} 0 & -1 & -(s+1) & -(s+2) \end{pmatrix}.$$

The Hirzebruch-surface block $H_n(s) \in M_4(\mathbb{Z})$ is

$$H_n(s) = \begin{pmatrix} 1 & 2 & 2s+n+4 & 2s+n+6 \\ 0 & 1 & 2s+n+2 & 2s+n+4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

With respect to $\mathcal{C}_{n,m}(s)$, the $(n+m+2) \times (n+m+2)$ matrix of the Euler form is

$$S_{n,m}(s) = \begin{pmatrix} T_n & 0 & B_n(s) \\ 0 & I_m & -\mathbf{1}_{m \times 4} \\ 0 & 0 & H_n(s) \end{pmatrix}.$$

Here $\mathbf{1}_{m \times 4}$ denotes the matrix all of whose entries are equal to 1.

Proof. On the local quotient chart $[\mathbb{A}^2/\mu_n]$, consider the standard μ_n -equivariant Koszul resolution [16, cf. Proposition 3.10]:

$$0 \longrightarrow \mathcal{O}_{\mathbb{A}^2} \otimes \rho_{i+2} \longrightarrow (\mathcal{O}_{\mathbb{A}^2} \otimes \rho_{i+1})^{\oplus 2} \longrightarrow \mathcal{O}_{\mathbb{A}^2} \otimes \rho_i \longrightarrow \mathcal{O}_0 \otimes \rho_i \longrightarrow 0.$$

Taking equivariant Homs to $\mathcal{O}_0 \otimes \rho_j$ gives

$$\chi(\mathcal{E}_i, \mathcal{E}_j) = \delta_{ij} - 2\delta_{j,i+1} + \delta_{j,i+2},$$

which is the block T_n displayed in (1). Although the characters are indexed cyclically, no wraparound term remains within the range $2 \leq i, j \leq n-1$; the indices in the displayed formula are therefore ordinary integers.

Let Ψ be the left adjoint of Φ . Specializing the Gugiatti–Rota adjoint formula [12, Theorem 3.1 and Example 3.5] gives

$$\Psi(\mathcal{E}_i) = 0 \quad (2 \leq i \leq n-3), \quad \Psi(\mathcal{E}_{n-1}) \simeq \mathcal{O}_E(-n+1).$$

For $n \geq 4$, the remaining relevant formula is

$$\Psi(\mathcal{E}_{n-2}) \simeq \mathcal{O}_E(-n)[1].$$

Indeed, $\rho_{\det} = \rho_2$ and $\rho_{n-2} = \rho_{\det}^\vee$, so the first case of [12, Theorem 3.1] gives $\mathcal{O}_E(E)[1]$, which is $\mathcal{O}_E(-n)[1]$ because $E^2 = -n$. When $n = 3$, the omitted source corresponds to the special representation ρ_1 .

The same divisor-duality computation as in Proposition 5.9 therefore gives, for a divisor D on $\mathbb{F}_n$,

$$\chi(\mathcal{E}_{n-2}, \Phi(\sigma^* \mathcal{O}(D))) = D \cdot C_0 + 1, \quad \chi(\mathcal{E}_{n-1}, \Phi(\sigma^* \mathcal{O}(D))) = -D \cdot C_0.$$

The first formula applies when $n \geq 4$. For the four line bundles in $\mathcal{C}_{n,m}(s)$, the intersection numbers are

$$D \cdot C_0 = 0, \quad 1, \quad s+1, \quad s+2.$$

These values give (2), including the stated truncation when $n = 3$. Since E is disjoint from $D_1, \dots, D_m$, the remaining local pairings satisfy

$$\chi(\mathcal{E}_i, \Phi(\mathcal{O}_{D_a}(-1))) = 0$$

for every i and a .

Because the blow-up centers are pairwise distinct, the curves $D_1, \dots, D_m$ are pairwise disjoint. Orlov's blow-up formula [22, Theorem 4.3 and Corollary 4.4] and the divisor-duality computation in Proposition 5.9 apply to each of them and give

$$\chi(\mathcal{O}_{D_a}(-1), \mathcal{O}_{D_b}(-1)) = \delta_{ab}, \quad \chi(\mathcal{O}_{D_a}(-1), \sigma^*L) = -1, \quad \chi(\sigma^*L, \mathcal{O}_{D_a}(-1)) = 0$$

for every line bundle L on $\mathbb{F}_n$. These pairings yield I_m , $-\mathbf{1}_{m \times 4}$, and the opposite zero block.

The last block follows from Riemann–Roch on $\mathbb{F}_n$. Using

$$K_{\mathbb{F}_n} = -2C_0 - (n+2)f$$

together with

$$\chi(\mathcal{O}(A), \mathcal{O}(B)) = 1 + \frac{1}{2}(B - A) \cdot (B - A - K_{\mathbb{F}_n}),$$

gives $H_n(s)$ for the four Hille–Perling line bundles. Combining these calculations with the Ishii–Ueda ordering [16, Theorem 1.4], which supplies the remaining lower-left zero blocks, proves the formula for $S_{n,m}(s)$. $\square$

For $n = 3$ and $m = 8$, one has $T_3 = (1)$ and $B_3(s) = (0, -1, -(s+1), -(s+2))$, while $H_3(s)$ is the 4×4 block in Proposition 5.9. Together with I_8 and $-\mathbf{1}_{8 \times 4}$, the formula recovers $S_{\mathcal{X}_{10}}(s)$.

In particular, Proposition 6.2 applies to the one-point Cavey–Prince cascades starting from $\mathbb{P}(1, 1, n)$ whenever the centers on the minimal resolution are pairwise distinct [6].

Concluding remarks

The marked models of the six rigid Corti–Heuberger bases turn abstract existence into explicit exceptional collections associated with chosen cascade presentations. The worked example X_{10} adds an Euler-form computation and untwisted KKS descent, while the Brauer calculation distinguishes the six-point cascade on the singular coarse-surface side. Three questions naturally follow. First, although Proposition 6.1 already gives the length formula for mixed baskets of $\frac{1}{n}(1, 1)$ -points with varying n , explicit marked models and closed Euler matrices for such baskets remain to be developed. The task is not merely to juxtapose the one-point matrices: one must choose a collection on the common resolution and compute its Euler pairings with each local block, taking account of the marked blow-up configuration. A natural testing ground is provided by the Cavey–Prince two-point models $X_{k_1+k_2} \subset \mathbb{P}(1, 1, k_1, k_2)$, with singularities of types $\frac{1}{k_1}(1, 1)$ and $\frac{1}{k_2}(1, 1)$ [6, Corollary 7.1]. Second, it remains to compare the collections of Theorem 1.3 arising from different marked presentations of the same surface, and to determine whether they belong to a single mutation orbit. Finally, one may ask whether twisted KKS descent can accommodate the nonzero Brauer class in the six-point cascade and, if so, how to describe its descended components explicitly. These questions concern the dependence on geometric choices and the passage to the coarse surface, rather than the existence of the stack collections established here.

A Brauer groups and obstructions in the Corti–Heuberger cascades

We prove [Theorem 1.4](#): the Brauer group vanishes in the cascades with $1 \leq k \leq 5$ singular points and is $\mathbb{Z}/3$ in the six-point cascade. The proof combines the cascade description of [Section 5](#), the cokernel criterion of [Corollary 4.5](#), and invariance under smooth-point blow-ups. The argument proceeds in five steps:

1. [Appendix A.1](#) translates the cokernel into lattice terms and settles the one- and two-point cases.
2. [Appendix A.2](#) proves invariance along a cascade.
3. [Appendix A.3](#) computes the vanishing Brauer groups for three through five points.
4. [Appendix A.4](#) identifies the six-point glue group and the resulting obstruction to untwisted adherent descent.
5. [Appendix A.5](#) assembles the results for every family, including the auxiliary higher-index families.

Let $E_1, \dots, E_k$ be the pairwise disjoint exceptional (-3) -curves of the minimal resolution $\tilde{X} \rightarrow X$, and put $\Lambda = \bigoplus_i \mathbb{Z}E_i$. In the notation of [Remark 4.4](#), the intersection map is

$$\theta : \text{Pic}(\tilde{X}) \longrightarrow \Lambda^\vee, \quad D \longmapsto (D \cdot E_1, \dots, D \cdot E_k).$$

Here $\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ has basis $e_1, \dots, e_k$ dual to the E_i ; in particular, $\theta(E_i) = -3e_i$. By [Corollary 4.5](#), $\text{Br}(X) \cong \text{coker}(\theta)$.

The role of the bases Z_k is important: their families each contain only one isomorphism class, not merely a general member with a convenient description [[7](#), Remark 7]. Thus a toric or k -gon model in a base family is the base itself. The four-point model used below is instead in $X_{4,4/3}$; its comparison with $Z_4 = X_{4,7/3}$ requires one smooth-point blow-up. No deformation-invariance assertion for the Brauer group is needed in these comparisons.

Bright’s exact sequence applies to arbitrary rational surface singularities [[4](#), Proposition 1], whereas the classification carried out in [[4](#), Section 3] concerns the classical singular del Pezzo case obtained by contracting (-2) -curves. In the Corti–Heuberger cascades the exceptional curves over the singular points have self-intersection -3 , so the calculations below are not contained in that classification.

A.1 Glue in the exceptional lattice

The intersection lattice $P = \text{Pic}(\tilde{X})$ is free and unimodular. Indeed, the rational-surface description used in [Corollary 2.6](#) reduces this to $\mathbb{P}^2$ and the Hirzebruch surfaces, whose intersection matrices have determinant ± 1 . Each point blow-up adds an orthogonal rank-one summand of square -1 . In particular, intersection identifies P with $P^\vee$.

Definition A.1. The *saturation* of the exceptional lattice in P is

$$\Lambda^{\text{sat}} = (\Lambda \otimes \mathbb{Q}) \cap \text{Pic}(\tilde{X}).$$

The lattice is *primitive* if P/Λ is torsion-free, equivalently if $\Lambda^{\text{sat}} = \Lambda$. The finite quotient $\Lambda^{\text{sat}}/\Lambda$ is its *glue group*. Using the intersection form to identify $\Lambda^\vee$ with a

lattice in $\Lambda \otimes \mathbb{Q}$, the *discriminant group*

$$A_\Lambda := \Lambda^\vee / \Lambda \cong (\mathbb{Z}/3)^k$$

carries the induced discriminant bilinear form

$$b_\Lambda : A_\Lambda \times A_\Lambda \longrightarrow \mathbb{Q}/\mathbb{Z}, \quad b_\Lambda(\bar{x}, \bar{y}) = x \cdot y \pmod{\mathbb{Z}}.$$

An *integral overlattice* of Λ is a finite-index extension inside $\Lambda \otimes \mathbb{Q}$ whose pairings are integral. It lies in $\Lambda^\vee$, and its quotient by Λ is totally isotropic for b_Λ , meaning that every pairing between elements of the quotient is zero. Conversely, the inverse image of such a subgroup is an integral overlattice. This is the bilinear version of the overlattice correspondence discussed in [21, Section 1.4]; the quadratic-form version for even lattices is not needed here. In particular, $\Lambda^{\text{sat}}/\Lambda$ is totally isotropic, since its representatives belong to the integral lattice P .

The next lemma identifies the Brauer group canonically with the dual of the glue group, making the failure of primitivity an explicit obstruction.

Lemma A.2 (The Brauer group and the glue group). *There are natural isomorphisms*

$$\text{Br}(X) \cong \text{coker}(\theta) \cong \text{Hom}(\Lambda^{\text{sat}}/\Lambda, \mathbb{Q}/\mathbb{Z}).$$

Thus $\text{Br}(X) = 0$ if and only if Λ is primitive in P . As abstract finite abelian groups, $\text{Br}(X)$ and $\Lambda^{\text{sat}}/\Lambda$ are isomorphic, but this last identification is not canonical.

Proof. Applying $\text{Hom}(-, \mathbb{Z})$ to $0 \rightarrow \Lambda \rightarrow P \rightarrow P/\Lambda \rightarrow 0$ gives an exact sequence

$$P^\vee \longrightarrow \Lambda^\vee \longrightarrow \text{Ext}^1(P/\Lambda, \mathbb{Z}) \longrightarrow 0,$$

because P is free. Under the unimodular identification $P \cong P^\vee$, the first map is θ . For any finitely generated abelian group G , the sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ gives $\text{Ext}^1(G, \mathbb{Z}) \cong \text{Hom}(G_{\text{tors}}, \mathbb{Q}/\mathbb{Z})$. Here $(P/\Lambda)_{\text{tors}} = \Lambda^{\text{sat}}/\Lambda$. The result follows from Corollary 4.5; a finite abelian group is noncanonically isomorphic to its character group. $\square$

This glue group is also the group used to define the defect invariant in [7, Section 2.1]. The following elementary restrictions explain why the one- and two-point cases need no choice of model.

Lemma A.3. *Every class in $\Lambda^{\text{sat}}/\Lambda$ can be represented in the form*

$$L = \frac{1}{3} \sum_{i=1}^k a_i E_i, \quad a_i \in \{0, \pm 1\}.$$

Such a representative satisfies the congruences

$$\sum_{i=1}^k a_i^2 \equiv 0 \pmod{3}, \quad \sum_{i=1}^k a_i \equiv 0 \pmod{3}.$$

Consequently:

- (i) *if $k = 1$ or $k = 2$, the lattice Λ is primitive;*

(ii) if $3 \leq k \leq 5$, every nonzero glue class is represented by

$$\pm \frac{E_a + E_b + E_c}{3}$$

for a triple $\{a, b, c\}$, and the glue subgroup $\Lambda^{\text{sat}}/\Lambda$ has order at most 3;

(iii) if $k = 6$, classes supported on three or on six exceptional curves are numerically possible. These congruences alone do not establish their existence; [Appendix A.4](#) shows that on Z_6 only nonzero glue classes with support six occur.

Proof. Since every class in $\text{Pic}(\tilde{X})$ has integral intersection with each E_i , the saturation Λ^{sat} is contained in $\Lambda^\vee$. The Gram matrix of Λ is $-3I_k$, so $A_\Lambda \cong (\mathbb{Z}/3)^k$, which gives the stated form of L . Integrality of the intersection form on $\text{Pic}(\tilde{X})$ gives

$$L^2 = -\frac{1}{3} \sum_i a_i^2 \in \mathbb{Z},$$

and hence $\sum_i a_i^2 \equiv 0 \pmod{3}$. Adjunction also gives

$$K_{\tilde{X}} \cdot E_i = 1.$$

It follows that

$$K_{\tilde{X}} \cdot L = \frac{1}{3} \sum_i a_i \in \mathbb{Z},$$

which proves the second congruence.

The first congruence says that the number of nonzero coefficients is divisible by 3. This proves (i). If $3 \leq k \leq 5$, a nonzero class has support of size 3, and the second congruence forces its three coefficients to have the same sign. By the cited overlattice correspondence, the glue subgroup is totally isotropic for b_Λ . After changing their signs, two support-three classes with distinct supports A and B have pairing

$$-\frac{\#(A \cap B)}{3} \pmod{\mathbb{Z}}.$$

For two distinct triples in a set of at most five elements, the intersection has size 1 or 2, so the two classes cannot be orthogonal. Hence all nonzero glue classes generate the same cyclic subgroup, proving (ii). For $k = 6$, supports of size 3 and 6 satisfy the first numerical condition, subject also to the stated congruence on the signs. In particular, the alternating coefficient vector on all six curves satisfies

$$\sum_{i=1}^6 a_i^2 = 6 \equiv 0 \pmod{3}, \quad \sum_{i=1}^6 a_i = 1 - 1 + 1 - 1 + 1 - 1 = 0.$$

Thus the numerical conditions permit the coefficient pattern $\frac{1}{3}(E_1 - E_2 + E_3 - E_4 + E_5 - E_6)$, as asserted in (iii). $\square$

Corollary A.4. *Let X be a log del Pezzo surface with at most two singular points, all of type $\frac{1}{3}(1, 1)$. Its Brauer group is trivial. In particular, $\text{Br}(Z_1) = \text{Br}(\mathbb{P}(1, 1, 3)) = 0$ and $\text{Br}(Z_2) = 0$.*

Proof. Combine [Lemmas A.2](#) and [A.3](#); the smooth case is the rational-surface vanishing used in [Remark 4.4](#). $\square$

For $k = 1$, this recovers the direct fiber-class computation in [Remark 4.4](#). For $3 \leq k \leq 5$, the same lemmas leave only the possibilities 0 and $\mathbb{Z}/3$. The explicit models below decide between them; the numerical congruences alone do not prove that an allowed glue class occurs.

A.2 Behavior under cascade blow-ups

The morphisms in a Corti–Heuberger cascade are blow-ups of nonsingular points by [7, Corollary 8]. Their effect on the intersection map is elementary.

Lemma A.5. *Let X be a normal projective surface with quotient singularities and rational minimal resolution, and let $X' \rightarrow X$ be the blow-up of a nonsingular point. This blow-up preserves the Brauer group:*

$$\mathrm{Br}(X') \cong \mathrm{Br}(X).$$

The same conclusion holds for a finite sequence of smooth-point blow-ups, including infinitely near centers.

Proof. On minimal resolutions, the induced morphism

$$\beta : \tilde{X}' \longrightarrow \tilde{X}$$

is the blow-up of a point disjoint from the exceptional curves $E_1, \dots, E_k$ of $\tilde{X} \rightarrow X$, since the resolution is an isomorphism over the smooth locus. The strict transforms of these curves give natural identifications

$$\Lambda_{X'} \cong \Lambda_X, \quad \Lambda_{X'}^\vee \cong \Lambda_X^\vee.$$

If D is the new exceptional curve, the Picard group decomposes as

$$\mathrm{Pic}(\tilde{X}') = \beta^* \mathrm{Pic}(\tilde{X}) \oplus \mathbb{Z}D, \quad D \cdot E_i = 0.$$

This decomposition implies that the intersection numbers satisfy

$$(\beta^*L + nD) \cdot E_i = L \cdot E_i$$

for every $L \in \mathrm{Pic}(\tilde{X})$ and every i . Under the displayed identification of the dual exceptional lattices, the image and cokernel of θ are therefore unchanged. The Brauer groups are these cokernels by [Corollary 4.5](#). The assertion for a sequence follows step by step, even when a center lies on a curve introduced by an earlier blow-up: that curve lies over the smooth locus, not over a singular point of the current surface. $\square$

Combining this lemma with [Section 5](#) gives, for every $X \in X_{k,d}$,

$$\mathrm{Br}(X) \cong \mathrm{Br}(Z_k). \tag{3}$$

It is therefore enough to compute the cokernel on one model in each base family, or on a surface connected to that base by smooth-point blow-ups. Uniqueness of the base, rather than a choice of a general member, justifies this reduction.

A.3 The cases $k = 3, 4, 5$

Three points: a toric model. The three-point base admits a direct toric calculation.

Corti–Heuberger exhibit polygon 20 as a toric qG -degeneration in the family $X_{3,5}$ [7, Section 7, Table 4 and Figure 11]. We check its singularities directly before using uniqueness of the base. Let $N \cong \mathbb{Z}^2$, and let $v_1, \dots, v_5 \in N$ be the primitive ray

generators of its fan, in counterclockwise order, with V_i the corresponding torus-invariant divisors:

$$v_1 = (1, 0), \quad v_2 = (-1, 3), \quad v_3 = (-2, 3), \quad v_4 = (-1, 0), \quad v_5 = (0, -1).$$

The first three two-dimensional cones have determinant 3, while the remaining two are smooth. The primitive rays inserted to resolve the three singular cones are

$$u_{12} = (0, 1), \quad u_{23} = (-1, 2), \quad u_{34} = (-1, 1).$$

Let E_1, E_2, E_3 be the corresponding exceptional torus-invariant curves. Each inserted ray is one third of the sum of its original neighbors, and the determinants of all adjacent rays in the subdivided fan equal one. Thus the subdivision is the minimal toric resolution, obtained by the ray-subdivision procedure of [9, Chapters 10–11]. Computing the boundary self-intersections from the fan relations as in [9, Theorem 10.4.4] gives

$$(V_1^2, E_1^2, V_2^2, E_2^2, V_3^2, E_3^2, V_4^2, V_5^2) = (0, -3, -1, -3, -1, -3, -1, 0).$$

The inserted curves are therefore precisely the exceptional (-3) -curves, so the surface has exactly three $\frac{1}{3}(1, 1)$ -points and no other singularities. After the unimodular change of coordinates $(x, y) \mapsto (x, x + y)$, these are the rays of the polygon cited above. It belongs to $X_{3,5}$ and hence is isomorphic to Z_3 by the uniqueness statement recalled in Section 5.

The boundary intersections give

$$\theta(V_1) = e_1, \quad \theta(V_2) = e_1 + e_2, \quad \theta(V_4) = e_3.$$

Consequently, the dual basis vectors satisfy

$$e_1 = \theta(V_1), \quad e_2 = \theta(V_2) - \theta(V_1), \quad e_3 = \theta(V_4),$$

so θ is surjective and the Brauer group vanishes:

$$\mathrm{Br}(Z_3) = \mathrm{Br}(X_{3,5}) = 0.$$

Equivalently, the divisor classes

$$\mathcal{O}(-V_1), \quad \mathcal{O}(V_1 - V_2), \quad \mathcal{O}(-V_4)$$

have intersection $-\delta_{ij}$ with E_j .

The k -gon construction. For the four- and five-point calculations we use the k -gon construction. It starts with a smooth del Pezzo surface of degree k containing a cyclic configuration $C_1 + \cdots + C_k$ of (-1) -curves, blows up the vertices $C_i \cap C_{i+1}$, and contracts the strict transforms of the C_i . Corti–Heuberger show that the resulting surface has k singularities of type $\frac{1}{3}(1, 1)$, degree $k/3$, and belongs to the family $X_{k,k/3}$ [7, Section 3.5, “ k -gons”]. Thus the models used below are respectively members of $X_{4,4/3}$ and $X_{5,5/3}$. Let D_i be the exceptional curve over the i -th vertex and E_i the strict transform of C_i , with cyclic indices. Direct intersection calculations give $E_i^2 = -1 - 2 = -3$, since each C_i passes through two vertices. The curves E_i are pairwise disjoint, and each D_i meets just E_i and E_{i+1} , transversally once. Consequently,

$$\theta(D_i) = e_i + e_{i+1}, \quad \theta(E_i) = -3e_i.$$

The degree also follows from the discrepancy computation in Corollary 3.6: the k blow-ups lower K^2 from k to zero, and contraction of the k disjoint (-3) -curves raises it to $k/3$.

Five points. For the five-point base, consider the degree-five del Pezzo surface $\mathrm{Bl}_{p_1, \dots, p_4} \mathbb{P}^2$, with exceptional classes Q_i and line class H . The curves

$$Q_1, \quad H - Q_1 - Q_2, \quad Q_2, \quad H - Q_2 - Q_3, \quad H - Q_1 - Q_4$$

have adjacent intersections one and nonadjacent intersections zero, so they form a cyclic pentagon of (-1) -curves. The construction above gives a member of $X_{5,5/3}$, which is the rigid base Z_5 by uniqueness. The five vectors $e_i + e_{i+1}$ are the columns of

$$A_5 = I + P_5,$$

where P_5 is the cyclic permutation matrix. A direct calculation gives

$$\det(A_5) = \prod_{\zeta^5=1} (1 + \zeta) = 2.$$

Consequently, $\mathbb{Z}^5/A_5\mathbb{Z}^5$ has order 2. Since the image of θ also contains $3\mathbb{Z}^5$, and multiplication by 3 is invertible on this order-two quotient, it follows that

$$A_5\mathbb{Z}^5 + 3\mathbb{Z}^5 = \mathbb{Z}^5.$$

It follows that θ is surjective and the Brauer group vanishes:

$$\mathrm{Br}(Z_5) = \mathrm{Br}(X_{5,5/3}) = 0.$$

Four points. For the four-point cascade, it is convenient to use a member of $X_{4,4/3}$, obtained from the rigid base $Z_4 = X_{4,7/3}$ by one blow-up at a nonsingular point [7, Corollary 8]. By Lemma A.5, it has the same cokernel as Z_4 . The vectors $e_i + e_{i+1}$ span a rank-three subgroup, so one additional divisor class is needed. To avoid confusing the exceptional classes on a degree-four del Pezzo surface with the (-3) -curves above, write

$$S = \mathrm{Bl}_{p_1, \dots, p_5} \mathbb{P}^2$$

with basis $H, Q_1, \dots, Q_5$, where $H^2 = 1$, $Q_i^2 = -1$, and all other pairings between basis elements vanish. Consider

$$\begin{aligned} C_1 &= Q_1, \\ C_2 &= H - Q_1 - Q_2, \\ C_3 &= Q_2, \\ C_4 &= 2H - Q_1 - Q_2 - Q_3 - Q_4 - Q_5. \end{aligned}$$

Their self-intersections are all -1 . The adjacent intersections are

$$C_1 \cdot C_2 = C_2 \cdot C_3 = C_3 \cdot C_4 = C_4 \cdot C_1 = 1,$$

and the nonadjacent intersections are

$$C_1 \cdot C_3 = 0, \quad C_2 \cdot C_4 = 2 - 1 - 1 = 0.$$

Their intersection graph is therefore a 4-cycle. In addition,

$$Q_3 \cdot C_1 = Q_3 \cdot C_2 = Q_3 \cdot C_3 = 0, \quad Q_3 \cdot C_4 = 1.$$

For general $p_1, \dots, p_5$, these intersections are transverse; in particular, Q_3 avoids the four vertices of the cycle. If π is their blow-up, the projection formula gives

$$\theta(\pi^*Q_3) = e_4.$$

Thus, we obtain

$$e_1 = \theta(D_4) - e_4,$$

$$e_2 = \theta(D_1) - e_1,$$

$$e_3 = \theta(D_2) - e_2.$$

It follows that θ is surjective and the 4-gon representative has trivial Brauer group. Since it is a one-point blow-up of Z_4 , [Lemma A.5](#) gives

$$\mathrm{Br}(Z_4) = \mathrm{Br}(X_{4,7/3}) \cong \mathrm{Br}(X_{4,4/3}) = 0.$$

A.4 The six-point obstruction

Unlike the five smaller cascades, the six-point cascade carries a nonzero Brauer obstruction. We first exhibit a relation among the toric boundary divisors that produces an order-three glue class in $\Lambda^{\mathrm{sat}}/\Lambda \subset A_\Lambda$. This proves nonvanishing, but does not yet determine the whole group. We then compute the full image of the intersection map: a single alternating-sum congruence modulo three describes it, showing that the Brauer group is exactly $\mathbb{Z}/3$, with no additional independent obstruction.

For the rigid base $Z_6 = X_{6,2}$ of [Section 5](#), Corti–Heuberger give the corresponding toric model in [7, Section 2.2.5]. We compute the intersection map by first describing the minimal toric resolution and then determining the lattice generated by the images of its boundary divisors. Let $N \cong \mathbb{Z}^2$, and let $v_1, \dots, v_6 \in N$ be the primitive ray generators of its fan, with V_i the corresponding torus-invariant divisors:

$$\begin{aligned} v_1 &= (1, 0), & v_2 &= (-1, 3), & v_3 &= (-2, 3), \\ v_4 &= (-1, 0), & v_5 &= (1, -3), & v_6 &= (2, -3), \end{aligned}$$

in counterclockwise order. The unimodular change $(x, y) \mapsto (x, x + y)$ sends these vertices to those of [7, Figure 1]. Indices are cyclic modulo 6 throughout. Every adjacent cone has determinant 3, and hence defines a cyclic quotient singularity of order 3. The primitive rays inserted in the minimal toric resolution are

$$u_i = \frac{v_i + v_{i+1}}{3}, \quad i \in \mathbb{Z}/6\mathbb{Z}.$$

The inserted rays are explicitly

$$\begin{aligned} u_1 &= (0, 1), & u_2 &= (-1, 2), & u_3 &= (-1, 1), \\ u_4 &= (0, -1), & u_5 &= (1, -2), & u_6 &= (1, -1). \end{aligned}$$

Let E_i be the exceptional torus-invariant curve corresponding to u_i . Since

$$\det(v_i, u_i) = \det(u_i, v_{i+1}) = 1,$$

the subdivision is smooth and gives the minimal toric resolution [9, Chapters 10–11]. The resolved boundary is the alternating cycle

$$V_1, E_1, V_2, E_2, \dots, V_6, E_6,$$

and the ray relations are

$$u_{i-1} + u_i = v_i, \quad v_i + v_{i+1} = 3u_i.$$

Therefore, [9, Theorem 10.4.4] gives

$$V_i^2 = -1, \quad E_i^2 = -3.$$

Consequently, the minimal resolution of each singularity consists of one (-3) -curve, so all six singularities are of type $\frac{1}{3}(1, 1)$.

A nonzero glue class. The obstruction comes from a divisibility relation. The principal-divisor formula $\text{div}(\chi^m) = \sum_\rho \langle m, v_\rho \rangle D_\rho$ of [9, Theorem 4.1.3], applied to the character $m = (0, 1)$, gives

$$\begin{aligned} 0 &= E_1 + 3V_2 + 2E_2 + 3V_3 + E_3 - E_4 \\ &\quad - 3V_5 - 2E_5 - 3V_6 - E_6. \end{aligned}$$

Rearranging gives

$$E_1 - E_2 + E_3 - E_4 + E_5 - E_6 = 3L, \tag{4}$$

where the divisor L is

$$L = -V_2 - V_3 + V_5 + V_6 - E_2 + E_5 \in \text{Pic}(\tilde{Z}_6).$$

Thus L lies in Λ^{sat} but not in Λ : the E_i are linearly independent, and its coefficients in this basis are $\pm 1/3$. Its class has order three and realizes the alternating pattern allowed by Lemma A.3. In particular, Lemma A.2 already implies that $\text{Br}(Z_6) \neq 0$. We next determine the whole group.

The cokernel. For a complete toric variety, the torus-invariant prime divisors generate $\text{Cl}(\tilde{Z}_6)$, and smoothness gives $\text{Cl}(\tilde{Z}_6) = \text{Pic}(\tilde{Z}_6)$ [9, Theorem 4.1.3]. Hence the toric boundary generates $\text{Pic}(\tilde{Z}_6)$, and its image under θ is generated by

$$\theta(E_i) = -3e_i, \quad \theta(V_j) = e_{j-1} + e_j,$$

with cyclic indices. Put $c_j = e_{j-1} + e_j$, where $e_0 = e_6$, and let $A \subset \mathbb{Z}^6$ be the subgroup generated by $c_2, \dots, c_6$. These five vectors, together with e_6 , form a basis of $\mathbb{Z}^6$: the matrix of their columns is lower triangular with diagonal entries one. Thus A is primitive of rank five. Moreover,

$$c_1 = c_2 - c_3 + c_4 - c_5 + c_6,$$

so A is also the span of all six images $\theta(V_j)$. Consider the surjective homomorphism

$$\lambda : \mathbb{Z}^6 \longrightarrow \mathbb{Z}, \quad (a_1, \dots, a_6) \longmapsto a_1 - a_2 + a_3 - a_4 + a_5 - a_6.$$

Every c_j lies in $\ker(\lambda)$. Since both A and $\ker(\lambda)$ have rank five and A is primitive, this inclusion is an equality. The image of the full toric boundary is therefore

$$\text{im}(\theta) = A + 3\mathbb{Z}^6 = \ker(\lambda) + 3\mathbb{Z}^6.$$

This sum is the kernel of the reduction of λ modulo 3. Indeed, if $\lambda(a) = 3q$, then $a - 3qe_1 \in \ker(\lambda)$, while the reverse inclusion is immediate. Consequently,

$$\text{im}(\theta) = \left\{ (a_1, \dots, a_6) \in \mathbb{Z}^6 : a_1 - a_2 + a_3 - a_4 + a_5 - a_6 \equiv 0 \pmod{3} \right\}.$$

The homomorphism $\lambda \bmod 3 : \mathbb{Z}^6 \rightarrow \mathbb{Z}/3$ is surjective, so the cokernel and Brauer group are

$$\text{coker}(\theta) \cong \mathbb{Z}/3, \quad \text{Br}(Z_6) = \text{Br}(X_{6,2}) \cong \mathbb{Z}/3.$$

By [Lemma A.2](#), the glue group also has order three. It is therefore generated by the class of L in (4); no support-three glue class occurs on Z_6 . Indeed, its two nonzero elements are $[L]$ and $-[L]$, whose coefficient vectors modulo three are $\pm(1, -1, 1, -1, 1, -1)$; both have all six entries nonzero. The congruences in [Lemma A.3](#) give necessary numerical conditions, not existence of every permitted support. Here the global toric divisor relations select the alternating support-six class, whereas the computations for $k = 3, 4, 5$ exclude even the numerically permitted support-three classes.

The obstruction to untwisted adherence. In particular, there cannot exist divisor classes L_i satisfying

$$L_i \cdot E_j = -\delta_{ij},$$

since such classes would make θ surjective.

By [18, Corollary 4.8], a semiorthogonal decomposition of the resolution whose components are untwisted adherent to the connected components of the exceptional locus would force $X_{6,2}$ to be torsion-free in the sense of [18, Definition 4.7], equivalently $\text{Br}(X_{6,2}) = 0$. The nonzero class computed above therefore prevents such an untwisted adherent decomposition. In the expected untwisted construction, each local algebra would be $K(3, 1)$. This does not rule out unrelated semiorthogonal decompositions or the twisted adherent constructions of KKS.

The resulting distinction is thus between the five smaller cascades, where the Brauer group vanishes, and the six-point cascade, where an order-three class obstructs untwisted adherent KKS descent. The auxiliary families $B_{1,16/3}$ and $B_{2,8/3}$ also have trivial Brauer group by [Corollary A.4](#). We finish by assembling these computations and making their propagation from the bases to every member of the cascades explicit.

A.5 Proof of [Theorem 1.4](#)

We first collect the computations on the bases. For $k = 1, 2$, [Corollary A.4](#) gives vanishing independently of the model. For $k = 3, 5$, [Appendix A.3](#) computes the Brauer group on a toric surface and a pentagon surface in the respective base families. Each is the corresponding Z_k , by uniqueness of the base in [Section 5](#). For $k = 4$, the same subsection uses a quadrilateral surface in $X_{4,4/3}$, not in the base family; its single smooth-point contraction to Z_4 and [Lemma A.5](#) give $\text{Br}(Z_4) = 0$. Finally, the toric computation in [Appendix A.4](#) gives $\text{Br}(Z_6) \cong \mathbb{Z}/3$. Altogether,

$$\text{Br}(Z_k) \cong \begin{cases} 0, & 1 \leq k \leq 5, \\ \mathbb{Z}/3, & k = 6. \end{cases}$$

For every surface X in an index-one family $X_{k,d}$, (3) now gives $\text{Br}(X) \cong \text{Br}(Z_k)$. This uses the smooth-point blow-up presentation of each surface, not an assumption that its Brauer group is determined by a general member.

The auxiliary families $B_{1,16/3}$ and $B_{2,8/3}$ are covered directly by [Corollary A.4](#), since their members have one and two singular points, respectively. They also fit the cascade argument: the alternative routes in [7, Remark 9], recalled in [Section 5](#), give a smooth-point blow-up $X' \rightarrow B$ and four floating contractions from X' to Z_k , for $k = 1, 2$

respectively. Reversing those contractions and applying [Lemma A.5](#) gives

$$\mathrm{Br}(B) \cong \mathrm{Br}(X') \cong \mathrm{Br}(Z_k) = 0.$$

Together with the already treated base Z_1 , these account for all higher-index families.

We have now covered every family in the six cascades. The final assertion follows from [18, Corollary 4.8]: the nonzero Brauer group in the six-point cascade obstructs an untwisted adherent decomposition, whereas this Brauer obstruction is absent in the other five cascades. This proves [Theorem 1.4](#). $\square$

B Explicit exceptional collections on the rigid cascade bases

This appendix proves [Proposition 5.4](#) by specifying the initial collections for all six cascades. The models of the singular surfaces come from Corti–Heuberger and [Appendix A](#); the blow-down sequences and the elementary transformation below make their smooth resolutions explicit. These constructions serve a different purpose from the exceptional-lattice computations in [Appendix A](#).

After fixing uniform notation, we treat the six bases in order. [Appendices B.1](#) and [B.2](#) begin with $\mathbb{F}_3$ and distinguish the incidence conditions that create a second quotient point. The three- and six-point cases are checked by toric contractions; the four-point case requires an elementary transformation, and the five-point case uses the pentagon construction. Each case ends with its ordered resolution and stack collections, and [Corollary B.3](#) propagates them through a chosen marked index-one cascade.

The following guide summarizes the geometric steps. All modifications are ordinary blow-ups at smooth points; “infinitely near” describes the location of a center relative to the starting surface Y_0 .

k	Y_0	Blow-ups and centers	Key geometric step
1	$\mathbb{F}_3$	None	Use the negative section already present.
2	$\mathbb{F}_3$	3: one point and two infinitely near centers	Create a second (-3) -curve from the first exceptional curve.
3	$\mathbb{F}_3$	4 toric: one fixed point and three infinitely near centers	Reverse four boundary contractions to a ruled surface.
4	$\mathbb{F}_2$	7: five distinct points and two infinitely near centers	Contract a floating curve via an elementary transformation.
5	$\mathbb{P}^2$	9: four plane points, then five vertices (four infinitely near)	Blow up a pentagon to separate five (-3) -curves.
6	$\mathbb{F}_0$	8 toric: four fixed points and four infinitely near centers	Contract to a hexagon, then to the square fan.

The centers and their order are specified in the individual cases below; the table records their incidence types, not interchangeable choices of points.

We use the following notation throughout. For a sequence of ordinary point blow-ups

$$Y_m \xrightarrow{\beta_m} Y_{m-1} \longrightarrow \cdots \xrightarrow{\beta_1} Y_0, \quad \sigma = \beta_1 \circ \cdots \circ \beta_m,$$

let $D_i \subset Y_i$ be the exceptional curve created by β_i , and set

$$G_i = L\beta_m^* \cdots L\beta_{i+1}^* \mathcal{O}_{D_i}(-1). \quad (5)$$

Thus $G_m = \mathcal{O}_{D_m}(-1)$. If $\langle L_1, \dots, L_t \rangle$ is a full exceptional collection of line bundles on Y_0 , [Theorem 2.5](#) gives the ordered full exceptional collection

$$\langle G_m, \dots, G_1, \sigma^* L_1, \dots, \sigma^* L_t \rangle. \quad (6)$$

In particular, infinitely near centers are allowed; the objects in (5) are not replaced by sheaves on the final strict transforms. The starting collections, recalled in [Section 2](#), are

$$\begin{aligned} \mathcal{B}_{\mathbb{P}^2} &= \langle \mathcal{O}, \mathcal{O}(H), \mathcal{O}(2H) \rangle, \\ \mathcal{B}_{\mathbb{F}_n} &= \langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + nf), \mathcal{O}(C_0 + (n+1)f) \rangle. \end{aligned}$$

Here H is a line class, and C_0, f are the section and fiber classes with $C_0^2 = -n$. We denote the collection on $\tilde{Z}_k$ by $\mathcal{E}_k$, the canonical stack of Z_k by $\mathcal{Z}_k$, and its Ishii–Ueda embedding by Φ_k . Applying a functor to a collection means applying it entry by entry, in the displayed order.

B.1 The one-point base

The one-point base supplies the starting ruled-surface collection without any additional blow-ups, and serves as the reference case for the next construction. The minimal resolution of $Z_1 = \mathbb{P}(1, 1, 3)$ is $\mathbb{F}_3$, with exceptional curve C_0 . No point blow-ups are needed, and we take

$$\mathcal{E}_1 = \langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + 3f), \mathcal{O}(C_0 + 4f) \rangle.$$

The corresponding full exceptional collection on the canonical stack is

$$\langle \mathcal{E}_p, \Phi_1(\mathcal{O}), \Phi_1(\mathcal{O}(f)), \Phi_1(\mathcal{O}(C_0 + 3f)), \Phi_1(\mathcal{O}(C_0 + 4f)) \rangle,$$

of length five, where $\mathcal{E}_p$ is the residual-gerbe object defined in [Theorem 1.1](#).

B.2 The two-point base

We construct a second quotient point while preserving the original negative section of $\mathbb{F}_3$, thereby specifying the incidence data needed for the two-point base collection. The two-point case of the directed MMP in [7, Section 6] has a branch

$$Z_2 = X_{2,17/3} \longrightarrow T \longrightarrow \mathbb{P}(1, 1, 3), \quad \text{Sing}(T) = \left\{ \frac{1}{3}(1, 1), A_1 \right\},$$

with contractions of types (E.4) and (E.2), respectively. We give a marked reconstruction of this branch and verify its identification with Z_2 independently of a numerical blow-up count.

Proposition B.1. *Choose $q_1 \in \mathbb{F}_3 \setminus C_0$, and let f_0 be its ruling fiber. Blow up q_1 , with exceptional curve D_1 , and then choose $q_2 \in D_1$ away from the strict transform of f_0 . Blow up q_2 , with exceptional curve D_2 . Finally, choose q_3 on the strict transform of D_1 , away from both D_2 and the strict transform of f_0 , and blow it up. Then*

$$\tilde{Z}_2 \simeq Y_3 = \text{Bl}_{q_3} \text{Bl}_{q_2} \text{Bl}_{q_1} \mathbb{F}_3.$$

The two curves contracted to the quotient singularities are the untouched C_0 and the final strict transform A of D_1 .

Proof. Write Y_i for the surface after the first i blow-ups and T_i for the total transform of the class of D_i on Y_3 . Then $T_i T_j = -\delta_{ij}$, and pullbacks from $\mathbb{F}_3$ are orthogonal to all T_i . On Y_2 the strict transform A_2 of D_1 has square -2 . The last center lies on A_2 , but on neither of its two neighboring curves. Consequently

$$A = T_1 - T_2 - T_3, \quad A^2 = -3, \quad C_0 A = 0.$$

The strict transform $F' = f - T_1$ of f_0 has square -1 ; it meets C_0 and A once each. The curves D_2, D_3 remain (-1) -curves, each meeting A once and disjoint from C_0 and F' . This records the incidence data, including the two distinct tangent directions over q_1 .

To check that contracting C_0 and A gives the required del Pezzo surface, consider the rational divisor

$$P = -K_{Y_3} - \frac{1}{3}(C_0 + A) = \frac{5}{3}C_0 + 5f - \frac{4}{3}T_1 - \frac{2}{3}T_2 - \frac{2}{3}T_3.$$

Using $C_0^2 = -3$, $C_0 f = 1$, $f^2 = 0$, and the orthogonal total-transform basis above, the full square expansion is

$$\begin{aligned} P^2 &= \left(\frac{5}{3}\right)^2 (-3) + 2 \left(\frac{5}{3}\right) 5 + 25 \cdot 0 - \left(\frac{4}{3}\right)^2 - 2 \left(\frac{2}{3}\right)^2 \\ &= -\frac{25}{3} + \frac{50}{3} - \frac{16}{9} - \frac{8}{9} = \frac{75-16-8}{9} = \frac{17}{3}. \end{aligned}$$

The same basis gives the individual pairings with the marked curves:

$$\begin{aligned} PC_0 &= \frac{5}{3}(-3) + 5 = 0, & PA &= \frac{4}{3} - \frac{2}{3} - \frac{2}{3} = 0, \\ PF' &= \frac{5}{3} - \frac{4}{3} = \frac{1}{3}, & PD_2 &= PD_3 = \frac{2}{3}. \end{aligned}$$

Every other irreducible curve is the strict transform of a fiber not through q_1 , or of a curve $B \sim aC_0 + bf$ with $a \geq 1$. Since $B \neq C_0$, nonnegative intersection of distinct irreducible curves gives $BC_0 = b - 3a \geq 0$, hence $b \geq 3a$. Let m_1, m_2, m_3 be its multiplicities at the successive centers. Intersection with f_0 gives $m_1 \leq a$, while nonnegative intersection of its final strict transform with $A = T_1 - T_2 - T_3$ gives $\tilde{B}A = m_1 - m_2 - m_3 \geq 0$, hence $m_2 + m_3 \leq m_1$. Consequently,

$$P \cdot \tilde{B} = \frac{5}{3}b - \frac{4}{3}m_1 - \frac{2}{3}(m_2 + m_3) \geq \frac{5}{3}b - 2a \geq 3a > 0.$$

On a fiber not through q_1 , the intersection is $5/3$. Thus P is nef and big, and its only null curves are C_0, A . The pair $(Y_3, (C_0 + A)/3)$ is klt, so the basepoint-free theorem [19, Section 3.2] makes a sufficiently divisible multiple of P semiample. Its associated contraction has precisely these two disjoint (-3) -curves as exceptional locus and gives a normal surface Z with two $\frac{1}{3}(1, 1)$ -points. The discrepancy formula identifies P with the pullback of the ample divisor $-K_Z$.

There is also no ambiguity about the Fano index. The divisor-class group of Z is the quotient of $\text{Pic}(Y_3)$ by C_0, A , so it is free with basis f, T_2, T_3 . In this quotient

$$-K_Z = 5f - 2T_2 - 2T_3$$

is primitive. The surface therefore has Fano index one, and [7, Theorem 4 and Remark 7] identifies it with Z_2 . Since its exceptional curves have square -3 , Y_3 is its minimal resolution. $\square$

This construction also explains the intermediate A_1 in the MMP. On Y_2 , contracting C_0 and A_2 gives T ; the same nef-divisor argument, using $-K_{Y_2} - C_0/3$, supplies this contraction. Contracting D_2 and then the resulting (-1) -curve D_1 returns $\mathbb{F}_3$. Thus the resolutions and the singular surfaces fit into the commutative diagram

$$\begin{array}{ccccc} Y_3 & \xrightarrow{\beta_3} & Y_2 & \xrightarrow{\beta_1 \circ \beta_2} & \mathbb{F}_3 \\ \downarrow & & \downarrow & & \downarrow \\ Z_2 & \longrightarrow & T & \longrightarrow & \mathbb{P}(1, 1, 3). \end{array}$$

Reading the upper row in reverse, the configurations over singular points evolve as

$$C_0(-3) \rightsquigarrow C_0(-3) \sqcup A_2(-2) \rightsquigarrow C_0(-3) \sqcup A(-3).$$

The original quotient point is untouched; the inverse MMP creates the second one. With $\sigma_2 : Y_3 \rightarrow \mathbb{F}_3$ the composite, the seven-object collection is

$$\mathcal{E}_2 = \langle G_3, G_2, G_1, \sigma_2^* \mathcal{O}, \sigma_2^* \mathcal{O}(f), \sigma_2^* \mathcal{O}(C_0 + 3f), \sigma_2^* \mathcal{O}(C_0 + 4f) \rangle.$$

The collection $\langle \mathcal{E}_{p_1}, \mathcal{E}_{p_2}, \Phi_2(\mathcal{E}_2) \rangle$ on Z_2 has length nine. A threefold blow-up of $\mathbb{F}_3$ can also occur in the one-point cascade, but here both later centers lie over q_1 on the first exceptional curve and create A . The number of blow-ups alone does not specify which curves are to be contracted or the resulting singularity basket.

B.3 The three-point base

For the three-point base, we turn the toric resolution of [Appendix A.3](#) into a marked four-step blow-up of $\mathbb{F}_3$, so that its initial collection is specified by boundary intersections. The fan of Z_3 has rays

$$v_1 = (1, 0), \quad v_2 = (-1, 3), \quad v_3 = (-2, 3), \quad v_4 = (-1, 0), \quad v_5 = (0, -1),$$

and its minimal resolution inserts

$$u_{12} = (0, 1), \quad u_{23} = (-1, 2), \quad u_{34} = (-1, 1).$$

In the boundary order $V_1, E_1, V_2, E_2, V_3, E_3, V_4, V_5$, the self-intersections are

$$(0, -3, -1, -3, -1, -3, -1, 0).$$

We contract, in this order, the successive images of V_4, V_3, E_3, E_2 . The following table lists the neighboring rays at the actual stage of each contraction:

curve	neighboring rays	ray relation
V_4	u_{34}, v_5	$v_4 = u_{34} + v_5$
V_3	u_{23}, u_{34}	$v_3 = u_{23} + u_{34}$
E_3	u_{23}, v_5	$u_{34} = u_{23} + v_5$
E_2	v_2, v_5	$u_{23} = v_2 + v_5$

In every row the determinant of the two neighbors is one. The ray-sum criterion therefore verifies that the curve being removed is a (-1) -curve and that its contraction is a smooth toric blow-down [9, Theorem 10.4.4]. The remaining fan has rays v_1, u_{12}, v_2, v_5 ,

with boundary squares $(0, -3, 0, 3)$; it is the fan of $\mathbb{F}_3$, with negative section corresponding to u_{12} .

Reversing these four contractions prescribes the centers: insert u_{23} in the cone $\langle v_2, v_5 \rangle$, then u_{34} in $\langle u_{23}, v_5 \rangle$, then v_3 in $\langle u_{23}, u_{34} \rangle$, and finally v_4 in $\langle u_{34}, v_5 \rangle$. Each center is the intersection of the two indicated boundary curves. This gives $\sigma_3 : \tilde{Z}_3 \rightarrow \mathbb{F}_3$ and the full exceptional collection

$$\mathcal{E}_3 = \langle G_4, G_3, G_2, G_1, \sigma_3^* \mathcal{O}, \sigma_3^* \mathcal{O}(f), \sigma_3^* \mathcal{O}(C_0 + 3f), \sigma_3^* \mathcal{O}(C_0 + 4f) \rangle.$$

The creation stages for G_1, G_2, G_3, G_4 correspond respectively to the rays u_{23}, u_{34}, v_3, v_4 ; in particular the first two objects are understood by derived pullback, not as sheaves on the final (-3) -curves. Prepending the three gerbe objects gives

$$\langle \mathcal{E}_{p_1}, \mathcal{E}_{p_2}, \mathcal{E}_{p_3}, \Phi_3(\mathcal{E}_3) \rangle$$

on Z_3 , of length eleven.

B.4 The four-point base

A geometric reconstruction is needed here because the quadrilateral model in [Appendix A.3](#) lies in $X_{4,4/3}$, not in the rigid base family $Z_4 = X_{4,7/3}$. The following proposition supplies the additional contraction and the change of ruled surface required for its marked blow-up presentation.

Proposition B.2. *Let $S = \text{Bl}_{p_1, \dots, p_5} \mathbb{P}^2$, with the points general, and let $H, Q_1, \dots, Q_5$ be its standard divisor classes. Set*

$$\begin{aligned} C_1 &= Q_1, & C_2 &= H - Q_1 - Q_2, \\ C_3 &= Q_2, & C_4 &= 2H - Q_1 - Q_2 - Q_3 - Q_4 - Q_5. \end{aligned}$$

Blow up the vertices $r_i = C_i \cap C_{i+1}$, cyclically, with exceptional curves R_i , obtaining Y . The strict transform of the line tangent at p_1 to the conic through the five points is a (-1) -curve

$$D = H - Q_1 - R_4.$$

Its contraction $Y \rightarrow Y'$ identifies Y' with $\tilde{Z}_4$. Moreover, an elementary transformation at r_4 gives an explicit seven-point blow-up presentation $\sigma_4 : \tilde{Z}_4 \rightarrow \mathbb{F}_2$.

Proof. All divisor classes on Y in the calculations below are total pullbacks, except where strict transforms are explicitly specified. The four (-1) -curves C_i form a cycle with distinct vertices, so their strict transforms

$$E_i = C_i - R_{i-1} - R_i$$

satisfy $E_i^2 = -3$ and $E_i E_j = 0$ for $i \neq j$. The contraction $Y \rightarrow X_{4,4/3}$ is the 4-gon construction of [7, Section 3.5].

The conic through five general points is smooth. Its tangent line at p_1 contains none of $p_2, \dots, p_5$, by Bézout, and meets Q_1 at r_4 after the first blow-up. This point is different from r_1 , the direction of the secant through p_2 . Thus D is indeed the strict transform of an irreducible smooth rational curve, rather than just a divisor class. In the orthogonal blow-up basis a direct calculation gives

$$D^2 = 1 - 1 - 1 = -1, \quad (-K_Y)D = 3 - 1 - 1 = 1,$$

and

$$\begin{aligned} DE_1 &= 1 - 1 = 0, & DE_2 &= 1 - 1 = 0, \\ DE_3 &= 0, & DE_4 &= 2 - 1 - 1 = 0. \end{aligned}$$

These distinct irreducible curves are therefore disjoint. The image of D on $X_{4,4/3}$ lies in the smooth locus and is a floating (-1) -curve. Its contraction is a log del Pezzo surface with four unchanged $\frac{1}{3}(1,1)$ -points and degree $7/3$, hence is Z_4 by the classification in [Section 5](#). Contracting D on Y leaves the four E_i unchanged and gives its minimal resolution Y' .

To identify the starting ruled surface, first blow up only p_1 . This gives $\mathbb{F}_1$, with negative section Q_1 and ruling class $H - Q_1$. The point r_4 is already specified on Q_1 by the tangent direction of the conic; its definition does not require blowing up the other four points first. Put $W = \text{Bl}_{r_4} \mathbb{F}_1$. The strict transform of that ruling fiber is D . Contracting it is the elementary transformation of the ruled surface at a point on its negative section. The ruling descends to a $\mathbb{P}^1$ -bundle over $\mathbb{P}^1$, and the image of the strict transform of Q_1 is a section of square -2 , since it is disjoint from D . The classification of ruled surfaces therefore identifies the new surface with $\mathbb{F}_2$ [[13](#), Chapter V, Section 2].

It remains to account for the marked centers, rather than just their number. On W , perform the blow-ups at p_2, p_3, p_4, p_5 , then at r_1, r_2, r_3 . All these centers, including the two directions r_2, r_3 on the exceptional curve over p_2 , lie away from the strict transform of D : the tangent line avoids the four points, and $r_1 \neq r_4$. These seven blow-ups commute with the contraction of D . More explicitly, the isomorphism

$$W \setminus D \simeq \mathbb{F}_2 \setminus \{t\},$$

where t is the image of D , transports these centers and their successive infinitely near directions to $\mathbb{F}_2$. Blow up the images of p_2, p_3, p_4, p_5 , followed by the transported r_1, r_2, r_3 . This is the asserted morphism σ_4 , and proves the isomorphism by commuting the actual modifications. $\square$

Using precisely this order of seven centers in [\(5\)](#), we obtain

$$\mathcal{E}_4 = \langle G_7, \dots, G_1, \sigma_4^* \mathcal{O}, \sigma_4^* \mathcal{O}(f), \sigma_4^* \mathcal{O}(C_0 + 2f), \sigma_4^* \mathcal{O}(C_0 + 3f) \rangle.$$

Here C_0 is the negative section on the new $\mathbb{F}_2$. The collection has eleven objects, and

$$\langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_4}, \Phi_4(\mathcal{E}_4) \rangle$$

has length fifteen on Z_4 . In this display the p_i label the quotient singularities, not the original plane blow-up centers.

B.5 The five-point base

The five-point base needs no further contraction beyond the pentagon construction: its nine blow-ups directly provide the marked plane model and hence the initial Orlov collection. On the degree-five del Pezzo surface $S_5 = \text{Bl}_{p_1, \dots, p_4} \mathbb{P}^2$, take the pentagon from [Appendix A.3](#):

$$\begin{aligned} C_1 &= Q_1, & C_2 &= H - Q_1 - Q_2, & C_3 &= Q_2, \\ C_4 &= H - Q_2 - Q_3, & C_5 &= H - Q_1 - Q_4. \end{aligned}$$

Each curve has square -1 , adjacent curves intersect once, and non-adjacent curves are disjoint. Blow up the five distinct vertices $r_i = C_i \cap C_{i+1}$, with exceptional curves R_i . Since exactly two vertices lie on each C_i , the strict transforms

$$E_i = C_i - R_{i-1} - R_i$$

have square $-1 - 2 = -3$. Blowing up the unique intersection of any adjacent pair separates it, so $E_i E_j = 0$ for $i \neq j$. By the 5-gon construction of [7, Section 3.5], their contraction belongs to $X_{5,5/3}$. Uniqueness of this base identifies the contraction with Z_5 , and consequently

$$\tilde{Z}_5 \simeq \text{Bl}_{r_1, \dots, r_5} \text{Bl}_{p_1, \dots, p_4} \mathbb{P}^2.$$

Here the vertex blow-ups are on S_5 , not at nine distinct points of the plane: r_1, r_5 lie over p_1 , r_2, r_3 lie over p_2 , and r_4 maps to a fifth distinct plane point. Order the centers as $p_1, \dots, p_4, r_1, \dots, r_5$, and write σ_5 for their composite. The twelve-object Orlov collection is

$$\mathcal{E}_5 = \langle G_9, \dots, G_1, \sigma_5^* \mathcal{O}, \sigma_5^* \mathcal{O}(H), \sigma_5^* \mathcal{O}(2H) \rangle.$$

The canonical-stack collection $\langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_5}, \Phi_5(\mathcal{E}_5) \rangle$ has length seventeen, with the gerbe indices again referring to the singular points of Z_5 .

B.6 The six-point base

For the six-point base, the alternating toric boundary gives a route through the degree-six del Pezzo surface to $\mathbb{F}_0$, making the twelve-object resolution collection explicit despite the coarse-surface Brauer obstruction. For the toric model of Z_6 in [Appendix A.4](#), the rays are

$$\begin{aligned} v_1 &= (1, 0), & v_2 &= (-1, 3), & v_3 &= (-2, 3), \\ v_4 &= (-1, 0), & v_5 &= (1, -3), & v_6 &= (2, -3), \end{aligned}$$

and the resolution inserts

$$\begin{aligned} u_1 &= (0, 1), & u_2 &= (-1, 2), & u_3 &= (-1, 1), \\ u_4 &= (0, -1), & u_5 &= (1, -2), & u_6 &= (1, -1). \end{aligned}$$

Its boundary is $V_1, E_1, \dots, V_6, E_6$, with $V_i^2 = -1$ and $E_i^2 = -3$. We first contract $V_1, \dots, V_6$, in that order. At every stage, the neighbors of the ray v_i being removed are still u_{i-1}, u_i , and

$$u_{i-1} + u_i = v_i, \quad \det(u_{i-1}, u_i) = 1.$$

Thus every contraction is a smooth toric blow-down. Equivalently, the V_i are pairwise disjoint, so their self-intersections remain -1 until they are contracted. After any initial set I of these contractions, the self-intersection of the image of E_j is

$$-3 + \#(I \cap \{j, j+1\}),$$

with cyclic indices. After all six, the fan consists of the u_i and every boundary curve has square -1 . This is the smooth toric del Pezzo surface of degree six: its anticanonical divisor intersects each invariant curve positively, and its square is six.

Next contract the opposite boundary curves with rays u_1 and u_4 . They are disjoint, and their ray relations are

$$u_1 = u_6 + u_2, \quad u_4 = u_3 + u_5, \quad \det(u_6, u_2) = \det(u_3, u_5) = 1.$$

Both are therefore (-1) -curves at the moment of contraction. The remaining rays are $u_2, u_3, -u_2, -u_3$, and $\det(u_2, u_3) = 1$. In this lattice basis the fan is the square fan of $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$; this checks the endpoint directly, not just its degree.

Reversing the sequence prescribes eight toric point blow-ups: first insert u_4 between u_3, u_5 , then u_1 between u_6, u_2 , and then insert $v_6, v_5, \dots, v_1$, each between u_{i-1}, u_i . The insertions of u_4, u_1, v_6, v_3 blow up the four original fixed points of $\mathbb{F}_0$; the remaining four centers are infinitely near those producing u_4 and u_1 . Let $\sigma_6 : \tilde{Z}_6 \rightarrow \mathbb{F}_0$ be their composite. With f_1, f_2 the two ruling classes on $\mathbb{F}_0$, the resulting twelve-object collection is

$$\mathcal{E}_6 = \langle G_8, \dots, G_1, \sigma_6^* \mathcal{O}, \sigma_6^* \mathcal{O}(f_1), \sigma_6^* \mathcal{O}(f_2), \sigma_6^* \mathcal{O}(f_1 + f_2) \rangle.$$

Prepending the six residual-gerbe objects gives $\langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_6}, \Phi_6(\mathcal{E}_6) \rangle$, of length eighteen on Z_6 . This stack collection is compatible with the nonzero Brauer group computed in Appendix A: that obstruction concerns untwisted adherent descent to the singular coarse surface, not existence of a full exceptional collection on its stack.

The six constructions establish Proposition 5.4. Together with the cascade procedure already described in Section 5, the following corollary proves Theorem 1.3. We use $A_j^{(k)}$ for arbitrary entries of a base collection, reserving G_i for blow-up objects.

Corollary B.3. *Let X belong to an index-one family $X_{k,d}$, and put $s = K_{\tilde{Z}_k}^2 - K_X^2$. Choose a marked Corti–Heuberger presentation by s smooth-point blow-ups, and let $\gamma : \tilde{X} \rightarrow \tilde{Z}_k$ be the induced map of minimal resolutions. Write the base collection constructed above as*

$$\mathcal{E}_k = \langle A_1^{(k)}, \dots, A_{N_k}^{(k)} \rangle, \quad (N_1, \dots, N_6) = (4, 7, 8, 11, 12, 12),$$

and let $H_1, \dots, H_s$ be the Orlov objects of the additional blow-ups, defined by derived pullback as in (5). Then

$$D^b(\mathrm{coh} \tilde{X}) = \langle H_s, \dots, H_1, L\gamma^* A_1^{(k)}, \dots, L\gamma^* A_{N_k}^{(k)} \rangle$$

is a full exceptional collection. If $\mathcal{X}$ is the canonical stack and Φ its Ishii–Ueda embedding, the ordered collection

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_k}, \Phi(H_s), \dots, \Phi(H_1), \Phi(L\gamma^* \mathcal{E}_k) \rangle$$

is full exceptional, of length $N_k + s + k = 12 - K_X^2 + 4k/3$.

Thus, once a marked cascade presentation is chosen, every entry and its position in the collection are specified by the six initial lists, derived pullback, and the residual-gerbe construction. No further existence choice of a collection on a rational resolution is required.

Acknowledgements

I am grateful to my advisor, Sergey Galkin, for his guidance, support, and many valuable discussions. I thank Nicolau Saldanha and Carlos Tomei for supporting my travel to the RGAS School 2026, where part of this work was further developed. I also thank Alexander Kuznetsov for a helpful conversation during the school and for pointing me to the work of Karmazyn–Kuznetsov–Shinder, which motivated the comparison with the singular surface in Section 4. I am grateful to Alexey Elagin for useful discussions

at IMPA about derived categories of singular varieties and for emphasizing the distinction between the categorical behavior of singular surfaces and that of their smooth canonical stacks. Finally, I thank Alessio Corti for helpful correspondence on the Corti–Heuberger cascades, and in particular for pointing out the relevance of the Reid–Suzuki construction to X_{10} .

Funding

The author was supported by the Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro (FAPERJ) through the Nota 10 Excellence Scholarship.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT (OpenAI) to assist with language editing, organization, and manuscript formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

- [1] M. Akhtar et al. “Mirror symmetry and the classification of orbifold del Pezzo surfaces.” In: *Proc. Amer. Math. Soc.* 144.2 (2016), pp. 513–527. DOI: [10.1090/proc/12876](https://doi.org/10.1090/proc/12876).
- [2] V. Alexeev and V. V. Nikulin. *Del Pezzo and K3 Surfaces*. The Mathematical Society of Japan, 2006. DOI: [10.2969/msjmemoirs/015010000](https://doi.org/10.2969/msjmemoirs/015010000).
- [3] A. A. Beilinson. “Coherent sheaves on $\mathbb{P}^n$ and problems of linear algebra.” In: *Funct. Anal. Appl.* 12.3 (July 1978), pp. 214–216. DOI: [10.1007/bf01681436](https://doi.org/10.1007/bf01681436).
- [4] M. Bright. “Brauer groups of singular del Pezzo surfaces.” In: *Michigan Math. J.* 62.3 (Sept. 2013), pp. 657–664. DOI: [10.1307/mmj/1378757892](https://doi.org/10.1307/mmj/1378757892).
- [5] M. Brion. “Introduction to actions of algebraic groups.” In: *Les cours du CIRM* 1.1 (2010), pp. 1–22. DOI: [10.5802/ccirm.1](https://doi.org/10.5802/ccirm.1).
- [6] D. Cavey and T. Prince. “Del Pezzo surfaces with a single $1/k(1, 1)$ singularity.” In: *J. Math. Soc. Japan* 72.2 (2020), pp. 465–505. DOI: [10.2969/jmsj/79337933](https://doi.org/10.2969/jmsj/79337933).
- [7] A. Corti and L. Heuberger. “Del Pezzo surfaces with $\frac{1}{3}(1, 1)$ points.” In: *Manuscripta Math.* 153.1-2 (July 2016), pp. 71–118. DOI: [10.1007/s00229-016-0870-y](https://doi.org/10.1007/s00229-016-0870-y).
- [8] D. A. Cox. “The homogeneous coordinate ring of a toric variety.” In: *J. Algebraic Geom.* 4.1 (1995), pp. 17–50.
- [9] D. A. Cox, J. B. Little, and H. K. Schenck. *Toric Varieties*. Vol. 124. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 2011. DOI: [10.1090/gsm/124](https://doi.org/10.1090/gsm/124).
- [10] I. Dolgachev. “Weighted projective varieties.” In: *Group Actions and Vector Fields*. Springer Berlin Heidelberg, 1982, pp. 34–71. DOI: [10.1007/bfb0101508](https://doi.org/10.1007/bfb0101508).

- [11] A. D. Elagin. “Exceptional sets on del Pezzo surfaces with one log-terminal singularity.” In: *Math. Notes* 82.1-2 (July 2007), pp. 33–46. DOI: [10.1134/S000143460707005x](https://doi.org/10.1134/S000143460707005x).
- [12] G. Gugiatti and F. Rota. “Full Exceptional Collections for Anticanonical Log del Pezzo Surfaces.” In: *Int. Math. Res. Not. IMRN* 2023.21 (Oct. 2023), pp. 18803–18855. DOI: [10.1093/imrn/rnad228](https://doi.org/10.1093/imrn/rnad228).
- [13] R. Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. New York: Springer-Verlag, 1977. DOI: [10.1007/978-1-4757-3849-0](https://doi.org/10.1007/978-1-4757-3849-0).
- [14] L. Hille and M. Perling. “Exceptional sequences of invertible sheaves on rational surfaces.” In: *Compos. Math.* 147.4 (Mar. 2011), pp. 1230–1280. DOI: [10.1112/S0010437X10005208](https://doi.org/10.1112/S0010437X10005208).
- [15] A. R. Iano-Fletcher. “Working with weighted complete intersections.” In: *Explicit Birational Geometry of 3-folds*. Cambridge University Press, July 2000, pp. 101–174. DOI: [10.1017/cbo9780511758942.005](https://doi.org/10.1017/cbo9780511758942.005).
- [16] A. Ishii and K. Ueda. “The special McKay correspondence and exceptional collections.” In: *Tohoku Math. J. (2)* 67.4 (Dec. 2015), pp. 585–609. DOI: [10.2748/tmj/1450798075](https://doi.org/10.2748/tmj/1450798075).
- [17] J. M. Johnson and J. Kollár. “Kähler-Einstein metrics on log del Pezzo surfaces in weighted projective 3-spaces.” In: *Ann. Inst. Fourier (Grenoble)* 51.1 (2001), pp. 69–79. DOI: [10.5802/aif.1815](https://doi.org/10.5802/aif.1815).
- [18] J. Karmazyn, A. Kuznetsov, and E. Shinder. “Derived categories of singular surfaces.” In: *J. Eur. Math. Soc. (JEMS)* 24.2 (Feb. 2022), pp. 461–526. DOI: [10.4171/jems/1106](https://doi.org/10.4171/jems/1106).
- [19] J. Kollár and S. Mori. *Birational Geometry of Algebraic Varieties*. Cambridge University Press, Sept. 1998. DOI: [10.1017/cbo9780511662560](https://doi.org/10.1017/cbo9780511662560).
- [20] A. Kuznetsov. “Semiorthogonal decompositions in algebraic geometry.” In: *Proceedings of the International Congress of Mathematicians—Seoul 2014. Vol. II*. Seoul: Kyung Moon Sa, 2014, pp. 635–660.
- [21] V. V. Nikulin. “Integral symmetric bilinear forms and some of their applications.” In: *Mathematics of the USSR-Izvestiya* 14.1 (1980), pp. 103–167. DOI: [10.1070/IM1980v014n01ABEH001060](https://doi.org/10.1070/IM1980v014n01ABEH001060).
- [22] D. O. Orlov. “Projective Bundles, Monoidal Transformations, and Derived Categories of Coherent Sheaves.” In: *Izv. Math.* 41.1 (Feb. 1993), pp. 133–141. DOI: [10.1070/im1993v041n01abeh002182](https://doi.org/10.1070/im1993v041n01abeh002182).
- [23] M. Reid and K. Suzuki. “Cascades of projections from log del Pezzo surfaces.” In: *Number Theory and Algebraic Geometry*. Cambridge University Press, Jan. 2004, pp. 227–249. DOI: [10.1017/cbo9780511734946.012](https://doi.org/10.1017/cbo9780511734946.012).
- [24] The Stacks Project Authors. *The Stacks Project*. 2026.
- [25] J. Wunram. “Reflexive modules on quotient surface singularities.” In: *Mathematische Annalen* 279.4 (1988), pp. 583–598. DOI: [10.1007/BF01458530](https://doi.org/10.1007/BF01458530).